

common type of pooling layer. In this example, we use a 2×2 *pooling kernel*,⁹ with a stride of 2 and no padding. Only the max input value in each receptive field makes it to the next layer, while the other inputs are dropped. For example, in the lower-left receptive field in **Figure 14-9**, the input values are 1, 5, 3, 2, so only the max value, 5, is propagated to the next layer. Because of the stride of 2, the output image has half the height and half the width of the input image (rounded down since we use no padding).

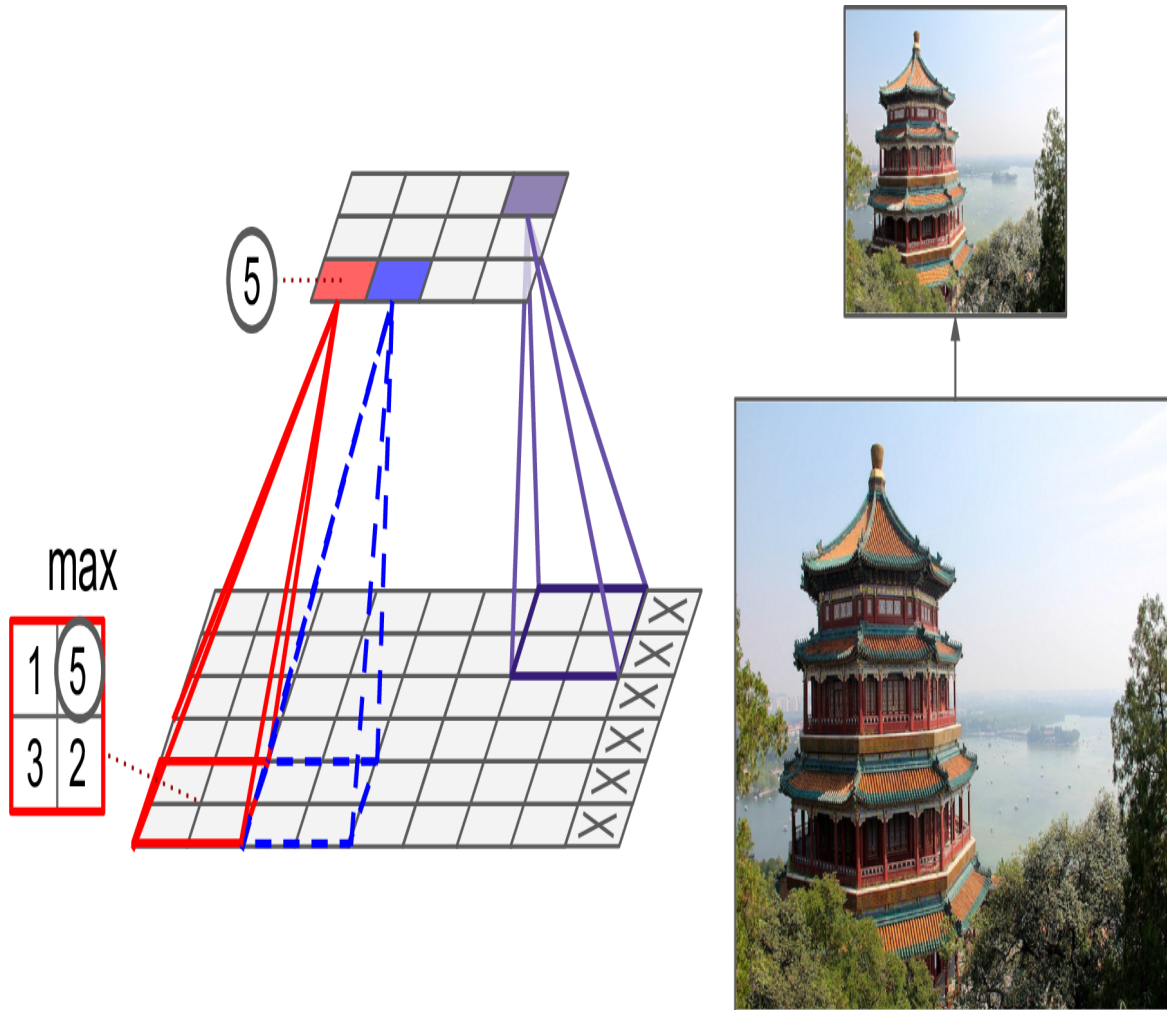


Figure 14-9. Max pooling layer (2×2 pooling kernel, stride 2, no padding)

NOTE

A pooling layer typically works on every input channel independently, so the output depth (i.e., the number of channels) is the same as the input depth.

Other than reducing computations, memory usage, and the number of parameters, a max pooling layer also introduces some level of *invariance* to small translations, as shown in **Figure 14-10**. Here we assume that the bright pixels have a lower value than dark pixels, and we consider three images (A, B, C) going through a max pooling layer with a 2×2 kernel and stride 2. Images B and C are the same as image A, but shifted by one and two pixels to the right. As you can see, the outputs of the max pooling layer for images A and B are identical. This is what translation invariance means. For image C, the output is different: it is shifted one pixel to the right (but there is still 50% invariance). By inserting a max pooling layer every few layers in a CNN, it is possible to get some level of translation invariance at a larger scale. Moreover, max pooling offers a small amount of rotational

invariance and a slight scale invariance. Such invariance (even if it is limited) can be useful in cases where the prediction should not depend on these details, such as in classification tasks.

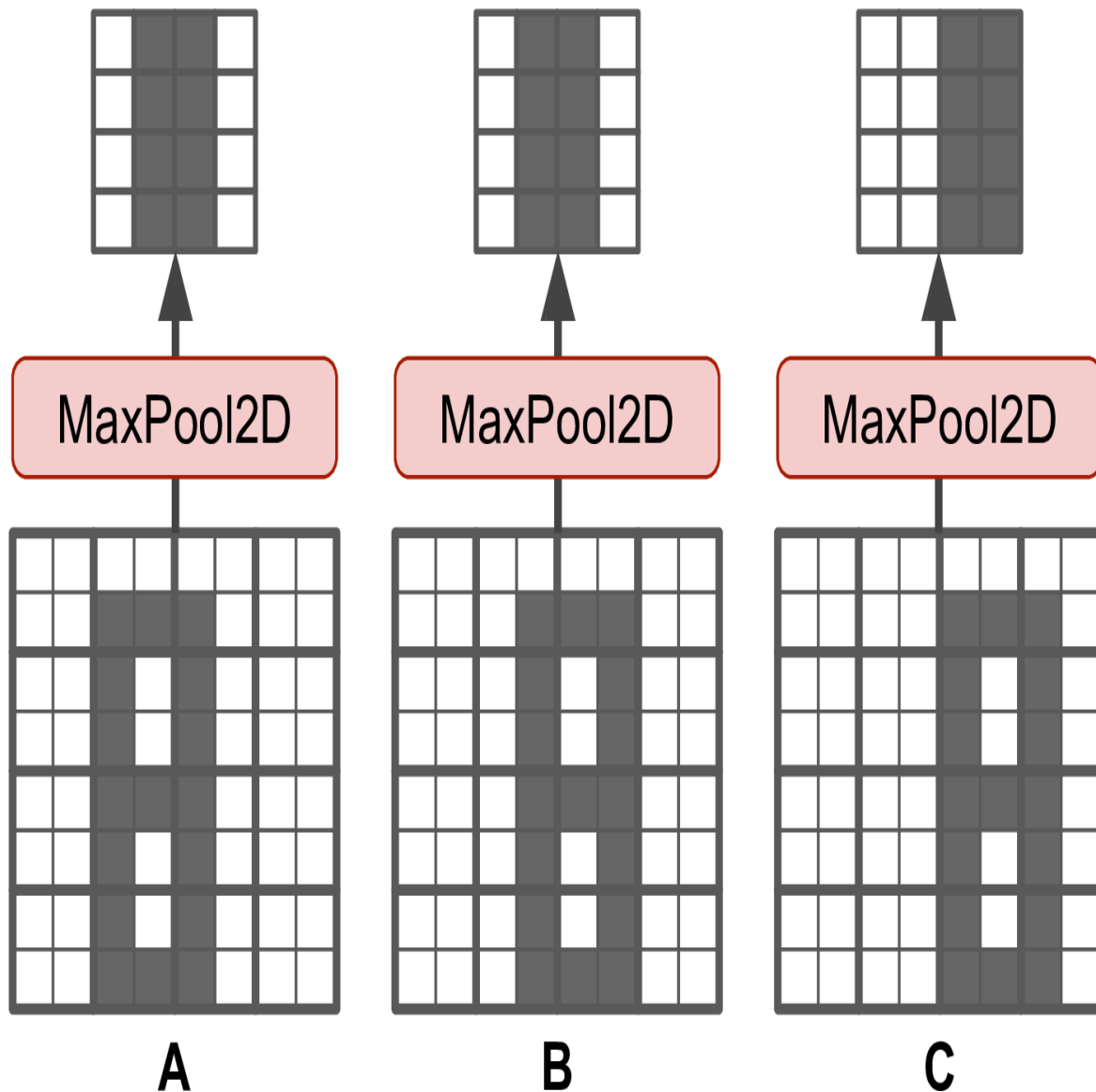


Figure 14-10. Invariance to small translations

However, max pooling has some downsides too. Firstly, it is obviously very destructive: even with a tiny 2×2 kernel and a stride of 2, the output will be two times smaller in both directions (so its area will be four times smaller), simply dropping 75% of the input values. And in some applications, invariance is not desirable. Take semantic segmentation (the task of classifying each pixel in an image according to the object that pixel belongs to, which we'll explore later in this chapter): obviously, if the input image is translated by one pixel to the right, the output should also be translated by one pixel to the right. The goal in this case is *equivariance*, not invariance: a small change to the inputs should lead to a corresponding small change in the output.

Implementing Pooling Layers With Keras

The following code creates a `MaxPooling2D` layer, alias `MaxPool2D`, using a 2×2 kernel. The strides default to the kernel size, so this layer will use a stride of 2 (both horizontally and vertically). By default, it uses “valid” padding (i.e., no padding at all):

```
max_pool = tf.keras.layers.MaxPool2D(pool_size=2)
```

To create an *average pooling layer*, just use `AveragePooling2D`, alias `AvgPool2D`, instead of `MaxPool2D`. As you might expect, it works exactly like a max pooling layer, except it computes the mean rather than the max. Average pooling layers used to be very popular, but people mostly use max pooling layers now, as they generally perform better. This may seem surprising, since computing the mean generally loses less information than computing the max. But on the other hand, max pooling preserves only the strongest features, getting rid of all the meaningless ones, so the next layers get a cleaner signal to work with. Moreover, max pooling offers stronger translation invariance than average pooling, and it requires slightly less compute.

Note that max pooling and average pooling can be performed along the depth dimension rather than the spatial dimensions, although this is not as common. This can allow the CNN to learn to be invariant to various features. For example, it could learn multiple filters, each detecting a different rotation of the same pattern (such as handwritten digits; see [Figure 14-11](#)), and the depthwise max pooling layer would ensure that the output is the same regardless of the rotation. The CNN could similarly learn to be invariant to anything else: thickness, brightness, skew, color, and so on.

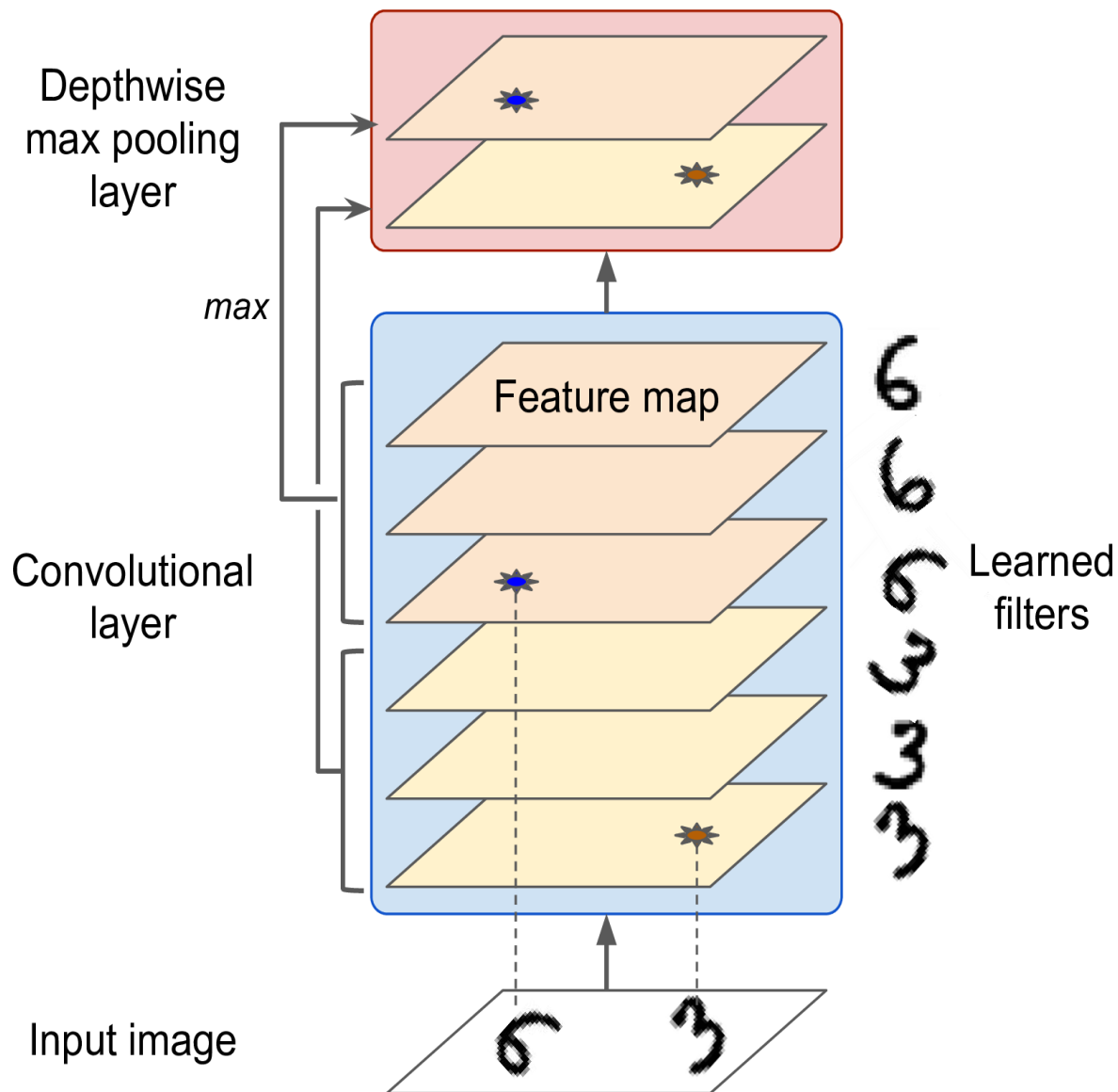


Figure 14-11. Depthwise max pooling can help the CNN learn to be invariant (to rotation in this case).

Keras does not include a depthwise max pooling layer, but it's not too difficult to implement a custom layer for that:

```
class DepthPool(tf.keras.layers.Layer):
    def __init__(self, pool_size=2, **kwargs):
        super().__init__(**kwargs)
        self.pool_size = pool_size

    def call(self, inputs):
        shape = tf.shape(inputs) # shape[-1] is the number of channels
        groups = shape[-1] // self.pool_size # number of channel groups
        new_shape = tf.concat([shape[:-1], [groups, self.pool_size]], axis=0)
        return tf.reduce_max(tf.reshape(inputs, new_shape), axis=-1)
```

This layer reshapes its inputs to split the channels into groups of the desired size (`pool_size`), then it uses `tf.reduce_max()` to compute the max of each group. This implementation assumes that the stride is equal to the pool size, which is generally what you want. Alternatively, you could use TensorFlow's `tf.nn.max_pool()` operation, and wrap in a `Lambda` layer to use it inside a Keras model, but sadly this op does not implement depthwise pooling for the GPU, only for the CPU.

One last type of pooling layer that you will often see in modern architectures is the *global average pooling layer*. It works very differently: all it does is compute the mean of each entire feature map (it's like an average pooling layer using a pooling kernel with the same spatial dimensions as the inputs). This means that it just outputs a single number per feature map and per instance. Although this is of course extremely destructive (most of the information in the feature map is lost), it can be useful just before the output layer, as we will see later in this chapter. To create such a layer, simply use the `GlobalAveragePooling2D` class, alias `GlobalAvgPool2D`:

```
global_avg_pool = tf.keras.layers.GlobalAvgPool2D()
```

It's equivalent to the following `Lambda` layer, which computes the mean over the spatial dimensions (height and width):

```
global_avg_pool = tf.keras.layers.Lambda(
    lambda X: tf.reduce_mean(X, axis=[1, 2]))
```

For example, if we apply this layer to the input images, we get the mean intensity of red, green, and blue for each image:

```
>>> global_avg_pool(images)
<tf.Tensor: shape=(2, 3), dtype=float32, numpy=
array([[0.64338624, 0.5971759, 0.5824972 ],
       [0.76306933, 0.26011038, 0.10849128]], dtype=float32)>
```

Now you know all the building blocks to create convolutional neural networks. Let's see how to assemble them.

CNN Architectures

Typical CNN architectures stack a few convolutional layers (each one generally followed by a ReLU layer), then a pooling layer, then another few convolutional layers (+ReLU), then another pooling layer, and so on. The image gets smaller and smaller as it progresses through the network, but it also typically gets deeper and deeper (i.e., with more feature maps), thanks to the convolutional layers (see [Figure 14-12](#)). At the top of the stack, a regular feedforward neural network is added, composed of a few fully connected layers (+ReLU), and the final layer outputs the prediction (e.g., a softmax layer that outputs estimated class probabilities).

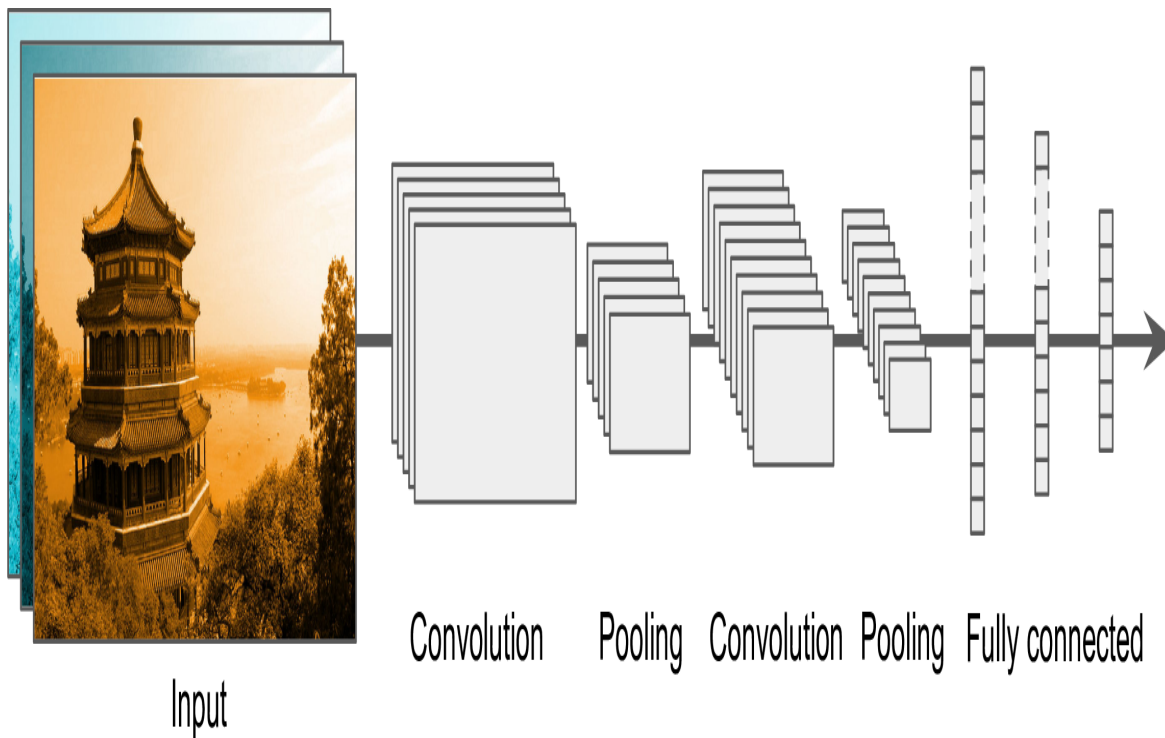


Figure 14-12. Typical CNN architecture

TIP

A common mistake is to use convolution kernels that are too large. For example, instead of using a convolutional layer with a 5×5 kernel, stack two layers with 3×3 kernels: it will use fewer parameters and require fewer computations, and it will usually perform better. One exception is for the first convolutional layer: it can typically have a large kernel (e.g., 5×5), usually with a stride of 2 or more: this will reduce the spatial dimension of the image without losing too much information, and since the input image only has three channels in general, it will not be too costly.

Here is how you can implement a basic CNN to tackle the Fashion MNIST dataset (introduced in [Chapter 10](#)):

```
from functools import partial

DefaultConv2D = partial(tf.keras.layers.Conv2D, kernel_size=3, padding="same",
                        activation="relu", kernel_initializer="he_normal")

model = tf.keras.Sequential([
    DefaultConv2D(filters=64, kernel_size=7, input_shape=[28, 28, 1]),
    tf.keras.layers.MaxPool2D(),
    DefaultConv2D(filters=128),
    DefaultConv2D(filters=128),
    tf.keras.layers.MaxPool2D(),
    DefaultConv2D(filters=256),
    DefaultConv2D(filters=256),
    tf.keras.layers.MaxPool2D(),
    tf.keras.layers.Flatten(),
    tf.keras.layers.Dense(units=128, activation="relu",
                          kernel_initializer="he_normal"),
    tf.keras.layers.Dropout(0.5),
    tf.keras.layers.Dense(units=64, activation="relu",
                          kernel_initializer="he_normal"),
    tf.keras.layers.Dropout(0.5),
    tf.keras.layers.Dense(units=10, activation="softmax")
])
```

Let's go through this code:

- We use the `functools.partial()` function (introduced in [Chapter 11](#)) to define `DefaultConv2D`, which acts just like `Conv2D` but with different default arguments: a small kernel size of 3, “same” padding, the ReLU activation function, and its corresponding He initializer.
- Next we create the `Sequential` model. Its first layer is a `DefaultConv2D` with 64 fairly large filters (7×7). It uses the default stride of 1 because the input images are not very large. It also sets `input_shape=[28, 28, 1]`, because the images are 28×28 pixels, with a single color channel (i.e., grayscale). When you load the Fashion MNIST dataset, make sure each image has this shape: you may need to use `np.reshape()` or `np.expanddims()` to add the channels dimension. Alternatively, you could use a `Reshape` layer as the first layer in the model.
- We then add a max pooling layer which uses the default pool size of 2, so it divides each spatial dimension by a factor of 2.
- Then we repeat the same structure twice: two convolutional layers followed by a max pooling layer. For larger images, we could repeat this structure several more times. The number of repetitions is a hyperparameter you can tune.
- Note that the number of filters doubles as we climb up the CNN toward the output layer (it is initially 64, then 128, then 256): it makes sense for it to grow, since the number of low-level features is often fairly low (e.g., small circles, horizontal lines), but there are many different ways to combine them into higher-level features. It is a common practice to double the number of filters after each pooling layer: since a pooling layer divides each spatial dimension by a factor of 2, we can afford to double the number of feature maps in the next layer without fear of exploding the number of parameters, memory usage, or computational load.
- Next is the fully connected network, composed of two hidden dense layers and a dense output layer. Since it’s a classification task with 10 classes, the output layer has 10 units and it uses the softmax activation function. Note that we must flatten the inputs just before the first dense layer, since it expects a 1D array of features for each instance. We also add two dropout layers, with a dropout rate of 50% each, to reduce overfitting.

If you compile this model using the “`sparse_categorical_crossentropy`” loss, and you fit the model to the Fashion MNIST training set, it should reach over 92% accuracy on the test set. It’s not state of the art, but it is pretty good, and clearly much better than what we achieved with dense networks in [Chapter 10](#).

Over the years, variants of this fundamental architecture have been developed, leading to amazing advances in the field. A good measure of this progress is the error rate in competitions such as the ILSVRC [ImageNet challenge](#). In this competition the top-five error rate for image classification fell from over 26% to less than 2.3% in just six years. The top-five error rate is the number of test images for which the system’s top five predictions did *not* include the correct answer. The images are fairly large (e.g., 256 pixels high) and there are 1,000 classes, some of which are really subtle (try distinguishing 120 dog breeds). Looking at the evolution of the winning entries is a good way to understand how CNNs work, and how research in Deep Learning progresses.

We will first look at the classical LeNet-5 architecture (1998), then several winners of the ILSVRC challenge: AlexNet (2012), GoogLeNet (2014), ResNet (2015), and SENet (2017). Along the way, we will also look at a few more architectures, including Xception, ResNext, DenseNet, MobileNet, CSPNet, and EfficientNet.

LeNet-5

The [LeNet-5 architecture](#)¹⁰ is perhaps the most widely known CNN architecture. As mentioned earlier, it was created by Yann LeCun in 1998 and has been widely used for handwritten digit recognition (MNIST). It is composed of the layers shown in [Table 14-1](#).

Table 14.1. LeNet-5 architecture

Layer	Type	Maps	Size	Kernel size	Stride	Activation
Out	Fully connected	–	10	–	–	RBF
F6	Fully connected	–	84	–	–	tanh
C5	Convolution	120	1×1	5×5	1	tanh
S4	Avg pooling	16	5×5	2×2	2	tanh
C3	Convolution	16	10×10	5×5	1	tanh
S2	Avg pooling	6	14×14	2×2	2	tanh
C1	Convolution	6	28×28	5×5	1	tanh
In	Input	1	32×32	–	–	–

As you can see, this looks pretty similar to our Fashion MNIST model: a stack of convolutional layers and pooling layers, followed by a dense network. Perhaps the main difference with more modern classification CNNs is the activation functions: today, we would use ReLU instead of tanh, and softmax instead of RBF. There were several other minor differences which don't really matter much, but in case you are interested, they are listed in the notebook. Yann LeCun's [website](#) also features great demos of LeNet-5 classifying digits.

AlexNet

The **AlexNet CNN architecture**¹¹ won the 2012 ImageNet ILSVRC challenge by a large margin: it achieved a top-five error rate of 17%, while the second best achieved only 26%! It was developed by Alex Krizhevsky (hence the name), Ilya Sutskever, and Geoffrey Hinton. It is similar to LeNet-5, only much larger and deeper, and it was the first to stack convolutional layers directly on top of one another, instead of stacking a pooling layer on top of each convolutional layer. **Table 14-2** presents this architecture.

Table 14. AlexNet architecture

Layer	Type	Maps	Size	Kernel size	Stride	Padding	Act
Out	Fully connected	–	1,000	–	–	–	Softmax
F10	Fully connected	–	4,096	–	–	–	ReLU
F9	Fully connected	–	4,096	–	–	–	ReLU
S8	Max pooling	256	6 × 6	3 × 3	2	valid	–
C7	Convolution	256	13 × 13	3 × 3	1	same	ReLU
C6	Convolution	384	13 × 13	3 × 3	1	same	ReLU
C5	Convolution	384	13 × 13	3 × 3	1	same	ReLU
S4	Max pooling	256	13 × 13	3 × 3	2	valid	–
C3	Convolution	256	27 × 27	5 × 5	1	same	ReLU
S2	Max pooling	96	27 × 27	3 × 3	2	valid	–
C1	Convolution	96	55 × 55	11 × 11	4	valid	ReLU
In	Input	3 (RGB)	227 × 227	–	–	–	–

To reduce overfitting, the authors used two regularization techniques. First, they applied dropout (introduced in [Chapter 11](#)) with a 50% dropout rate during training to the outputs of layers F9 and F10. Second, they performed *data augmentation* by randomly shifting the training images by various offsets, flipping them horizontally, and changing the lighting conditions.

DATA AUGMENTATION

Data augmentation artificially increases the size of the training set by generating many realistic variants of each training instance. This reduces overfitting, making this a regularization technique. The generated instances should be as realistic as possible: ideally, given an image from the augmented training set, a human should not be able to tell whether it was augmented or not. Simply adding white noise will not help; the modifications should be learnable (white noise is not).

For example, you can slightly shift, rotate, and resize every picture in the training set by various amounts and add the resulting pictures to the training set (see [Figure 14-13](#)). To do this, you can use Keras's data augmentation layers, introduced in [Chapter 13](#) (e.g., `RandomCrop`, `RandomRotation`, etc.). This forces the model to be more tolerant to variations in the position, orientation, and size of the objects in the pictures. To produce a model that's more tolerant of different lighting conditions, you can similarly generate many images with various contrasts. In general, you can also flip the pictures horizontally (except for text, and other asymmetrical objects). By combining these transformations, you can greatly increase the size of your training set.

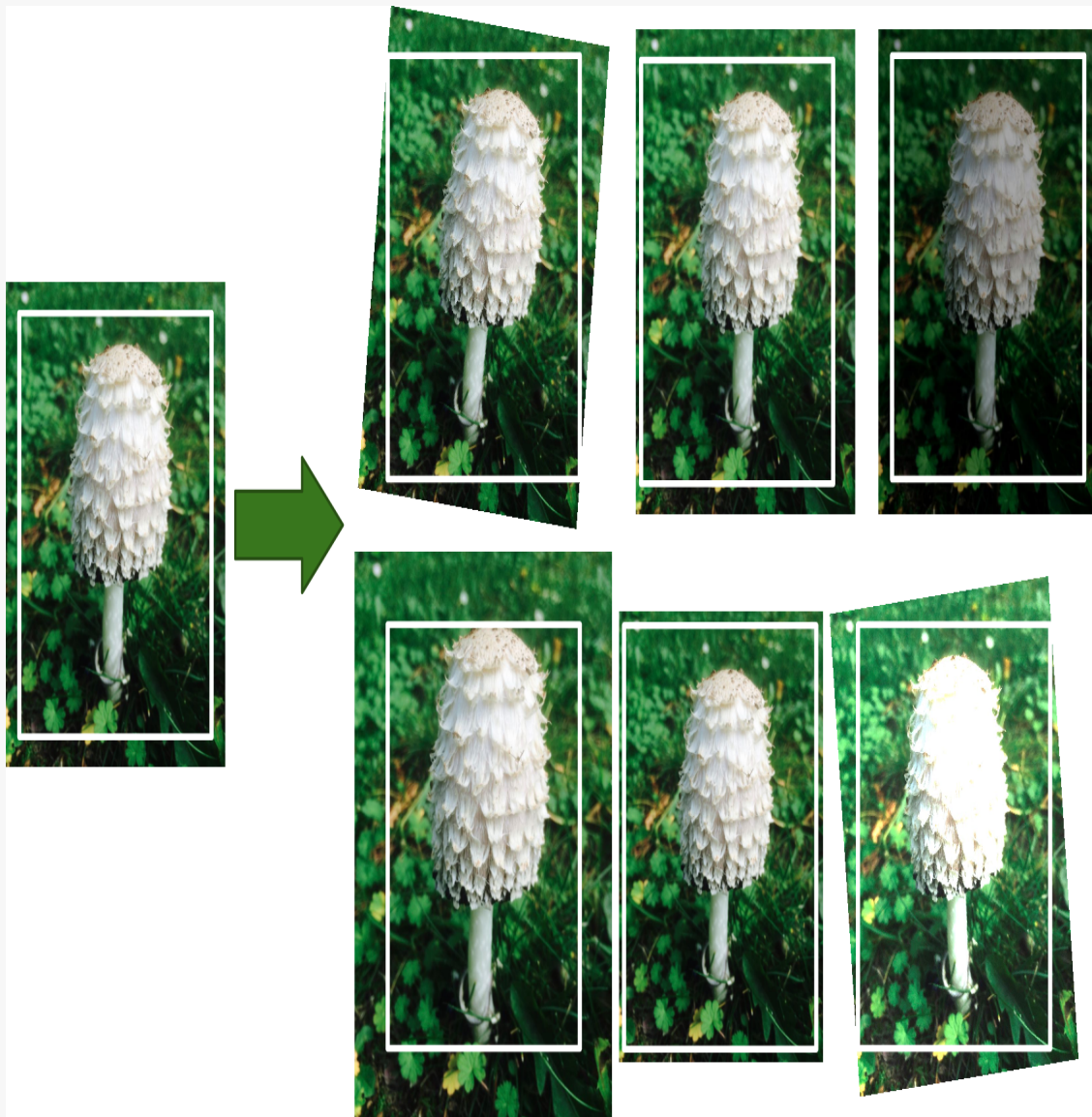


Figure 14-13. Generating new training instances from existing ones

Data augmentation is also useful when you have an unbalanced dataset: you can use it to generate more samples of the less frequent classes. This is called the *Synthetic Minority Oversampling Technique*, or SMOTE for short.

AlexNet also uses a competitive normalization step immediately after the ReLU step of layers C1 and C3, called *local response normalization* (LRN): the most strongly activated neurons inhibit other neurons located at the same position in neighboring feature maps. Such competitive activation has been observed in biological neurons. This encourages different feature maps to specialize, pushing them apart and forcing them to explore a wider range of features, ultimately improving generalization. Equation 14-2 shows how to apply LRN.

Equation 14-2. Local response normalization (LRN)

$$b_i = a_i \left(k + \alpha \sum_{j=j_{\text{low}}}^{j_{\text{high}}} a_j^2 \right)^{-\beta} \quad \text{with} \quad \begin{cases} j_{\text{high}} = \min \left(i + \frac{r}{2}, f_n - 1 \right) \\ j_{\text{low}} = \max \left(0, i - \frac{r}{2} \right) \end{cases}$$

In this equation:

- b_i is the normalized output of the neuron located in feature map i , at some row u and column v (note that in this equation we consider only neurons located at this row and column, so u and v are not shown).
- a_i is the activation of that neuron after the ReLU step, but before normalization.
- k , α , β , and r are hyperparameters. k is called the *bias*, and r is called the *depth radius*.
- f_n is the number of feature maps.

For example, if $r = 2$ and a neuron has a strong activation, it will inhibit the activation of the neurons located in the feature maps immediately above and below its own.

In AlexNet, the hyperparameters are set as follows: $r = 5$, $\alpha = 0.0001$, $\beta = 0.75$, and $k = 2$. This step can be implemented using the `tf.nn.local_response_normalization()` function (which you can wrap in a Lambda layer if you want to use it in a Keras model).

A variant of AlexNet called *ZF Net*¹² was developed by Matthew Zeiler and Rob Fergus and won the 2013 ILSVRC challenge. It is essentially AlexNet with a few tweaked hyperparameters (number of feature maps, kernel size, stride, etc.).

GoogLeNet

The *GoogLeNet architecture* was developed by Christian Szegedy et al. from Google Research,¹³ and it won the ILSVRC 2014 challenge by pushing the top-five error rate below 7%. This great performance came in large part from the fact that the network was much deeper than previous CNNs (as you'll see in Figure 14-15). This was made possible by subnetworks called *inception modules*,¹⁴ which allow GoogLeNet to use parameters much more efficiently than previous architectures: GoogLeNet actually has 10 times fewer parameters than AlexNet (roughly 6 million instead of 60 million).

Figure 14-14 shows the architecture of an inception module. The notation " $3 \times 3 + 1(S)$ " means that the layer uses a 3×3 kernel, stride 1, and "same" padding. The input signal is first fed to four different layers in parallel. All convolutional layers use the ReLU activation function. Note that top convolutional layers use different kernel sizes (1×1 , 3×3 , and 5×5), allowing them to capture patterns at different scales. Also note that every single layer uses a stride of 1 and "same" padding (even the max pooling layer), so their outputs all have the same height and width as their inputs. This makes it possible to concatenate all the outputs along the depth dimension in the final *depth concatenation layer* (i.e., stack the feature maps from all four top convolutional layers). It can be implemented using Keras's `Concatenate` layer, using the default `axis=-1`.

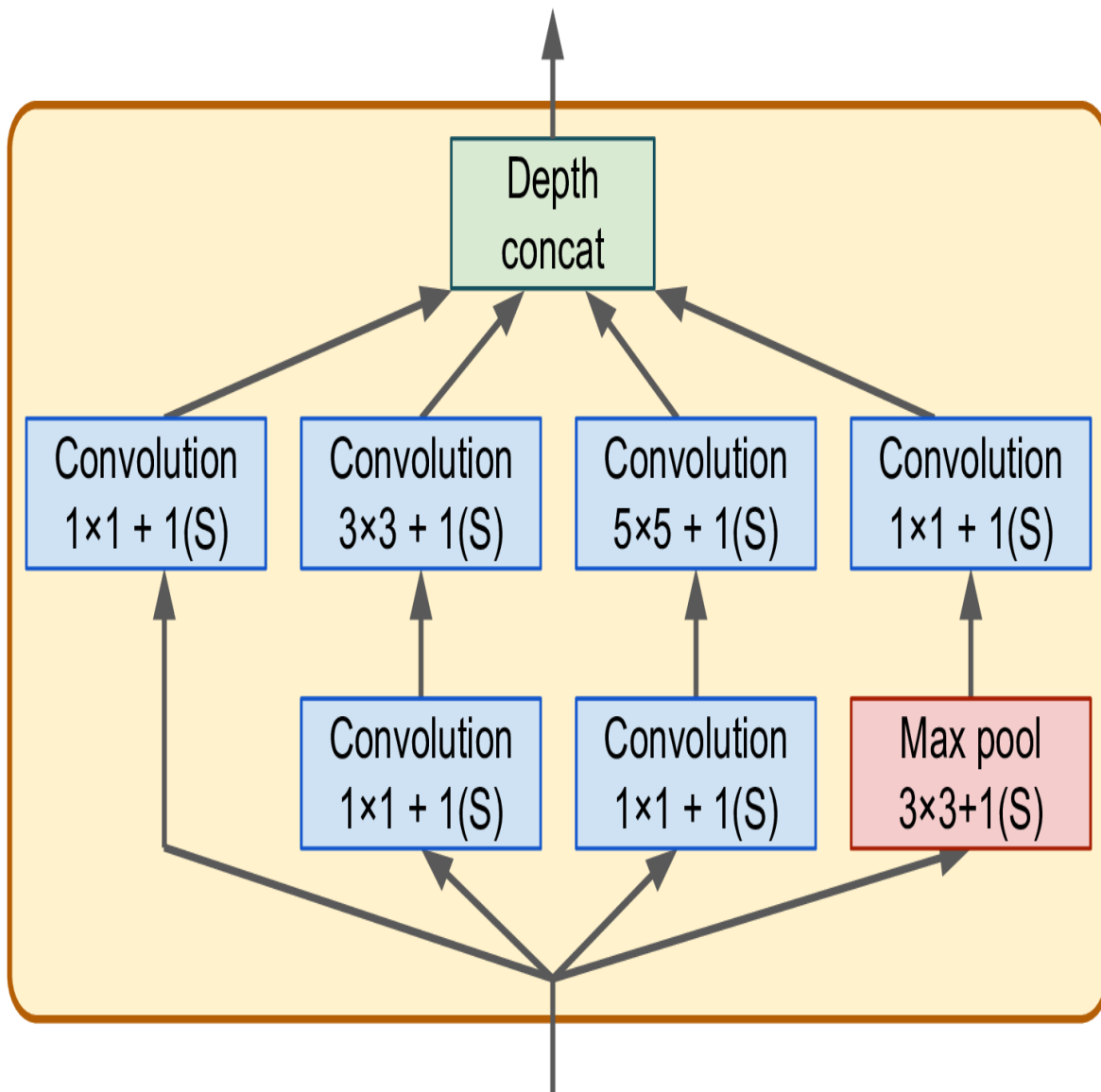


Figure 14-14. Inception module

You may wonder why inception modules have convolutional layers with 1×1 kernels. Surely these layers cannot capture any features because they look at only one pixel at a time, right? In fact, these layers serve three purposes:

- Although they cannot capture spatial patterns, they can capture patterns along the depth dimension (i.e., across channels).
- They are configured to output fewer feature maps than their inputs, so they serve as *bottleneck layers*, meaning they reduce dimensionality. This cuts the computational cost and the number of parameters, speeding up training and improving generalization.
- Each pair of convolutional layers ($[1 \times 1, 3 \times 3]$ and $[1 \times 1, 5 \times 5]$) acts like a single powerful convolutional layer, capable of capturing more complex patterns. Indeed, a convolutional layer is equivalent to sweeping a dense layer across the image (at each location, it only looks at a small receptive field), and these pairs of convolutional layers are equivalent to sweeping two-layer neural networks across the image.

In short, you can think of the whole inception module as a convolutional layer on steroids, able to output feature maps that capture complex patterns at various scales.

Now let's look at the architecture of the GoogLeNet CNN (see Figure 14-15). The number of feature maps output by each convolutional layer and each pooling layer is shown before the kernel size. The architecture is so deep that it has to be represented in three columns, but GoogLeNet is actually one tall stack, including nine inception modules (the boxes with the spinning tops). The six numbers in the inception modules represent the number of feature maps output by each convolutional layer in the module (in the same order as in Figure 14-14). Note that all the convolutional layers use the ReLU activation function.

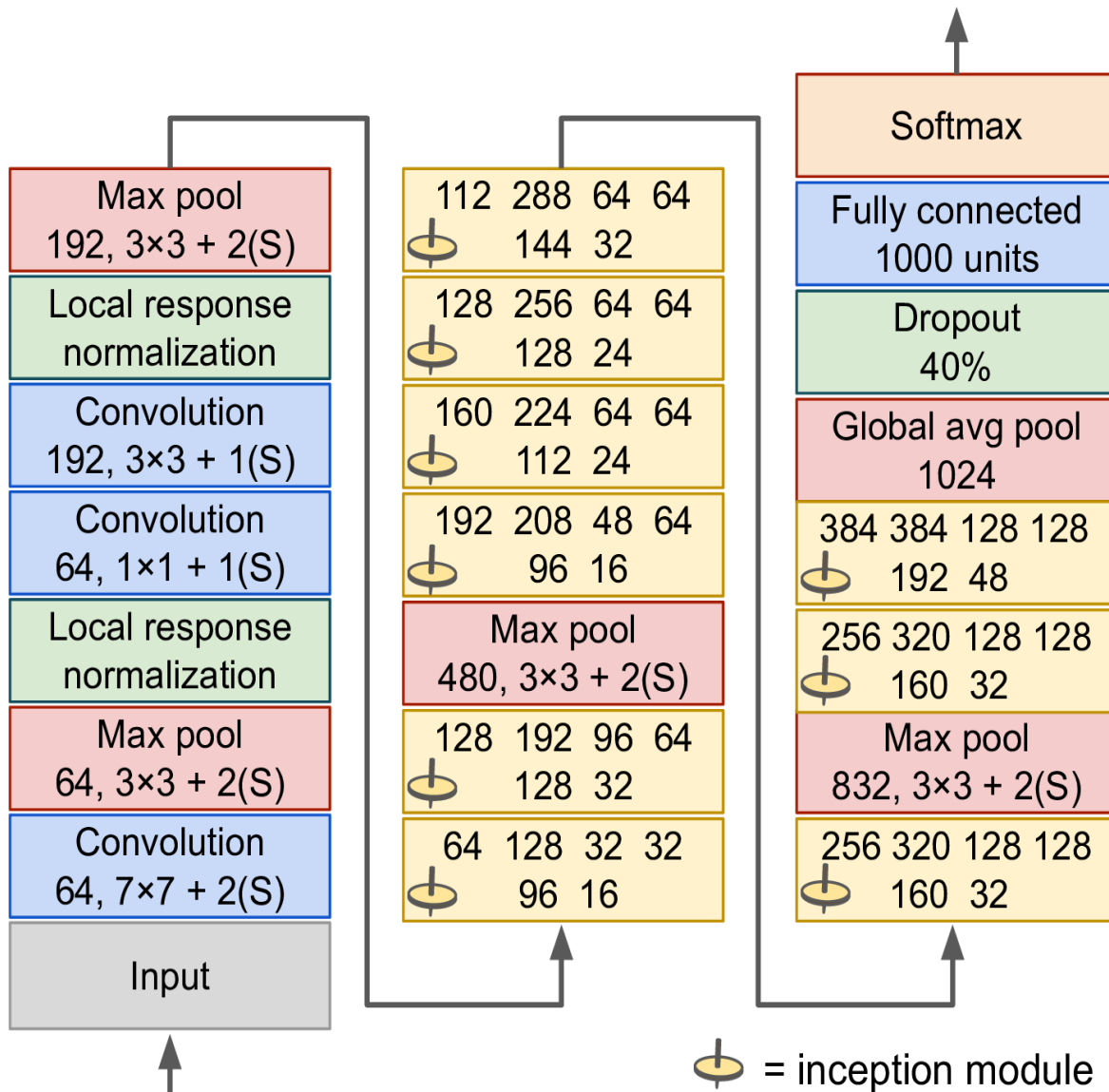


Figure 14-15. GoogLeNet architecture

Let's go through this network:

- The first two layers divide the image's height and width by 4 (so its area is divided by 16), to reduce the computational load. The first layer uses a large kernel size, 7×7 , so that much of the information is preserved.
- Then the local response normalization layer ensures that the previous layers learn a wide variety of features (as discussed earlier).

- Two convolutional layers follow, where the first acts like a bottleneck layer. As explained earlier, you can think of this pair as a single smarter convolutional layer.
- Again, a local response normalization layer ensures that the previous layers capture a wide variety of patterns.
- Next, a max pooling layer reduces the image height and width by 2, again to speed up computations.
- Then comes the CNN's *backbone*: a tall stack of nine inception modules, interleaved with a couple max pooling layers to reduce dimensionality and speed up the net.
- Next, the global average pooling layer outputs the mean of each feature map: this drops any remaining spatial information, which is fine because there was not much spatial information left at that point. Indeed, GoogLeNet input images are typically expected to be 224×224 pixels, so after 5 max pooling layers, each dividing the height and width by 2, the feature maps are down to 7×7 . Moreover, it is a classification task, not localization, so it does not matter where the object is. Thanks to the dimensionality reduction brought by this layer, there is no need to have several fully connected layers at the top of the CNN (like in AlexNet), and this considerably reduces the number of parameters in the network and limits the risk of overfitting.
- The last layers are self-explanatory: dropout for regularization, then a fully connected layer with 1,000 units (since there are 1,000 classes) and a softmax activation function to output estimated class probabilities.

The original GoogLeNet architecture also included two auxiliary classifiers plugged on top of the third and sixth inception modules. They were both composed of one average pooling layer, one convolutional layer, two fully connected layers, and a softmax activation layer. During training, their loss (scaled down by 70%) was added to the overall loss. The goal was to fight the vanishing gradients problem and regularize the network. However, it was later shown that their effect was relatively minor.

Several variants of the GoogLeNet architecture were later proposed by Google researchers, including Inception-v3 and Inception-v4, using slightly different inception modules to reach even better performance.

VGGNet

The runner-up in the ILSVRC 2014 challenge was **VGGNet**,¹⁵ developed by Karen Simonyan and Andrew Zisserman from the Visual Geometry Group (VGG) research lab at Oxford University. It had a very simple and classical architecture, with 2 or 3 convolutional layers and a pooling layer, then again 2 or 3 convolutional layers and a pooling layer, and so on (reaching a total of just 16 or 19 convolutional layers, depending on the VGG variant), plus a final dense network with 2 hidden layers and the output layer. It used only small 3×3 filters, but it had many of them.

ResNet

Kaiming He et al. won the ILSVRC 2015 challenge using a **Residual Network (or ResNet)**,¹⁶ that delivered an astounding top-five error rate under 3.6%. The winning variant used an extremely deep CNN composed of 152 layers (other variants had 34, 50, and 101 layers). It confirmed the general trend: computer vision models were getting deeper and deeper, with fewer and fewer parameters. The key to being able to train such a deep network is to use *skip connections* (also called *shortcut connections*): the signal feeding into a layer is also added to the output of a layer located a bit higher up the stack. Let's see why this is useful.

When training a neural network, the goal is to make it model a target function $h(\mathbf{x})$. If you add the input $\mathbf{x}$ to the output of the network (i.e., you add a skip connection), then the network will be forced to model $f(\mathbf{x}) = h(\mathbf{x}) - \mathbf{x}$ rather than $h(\mathbf{x})$. This is called *residual learning* (see **Figure 14-16**).

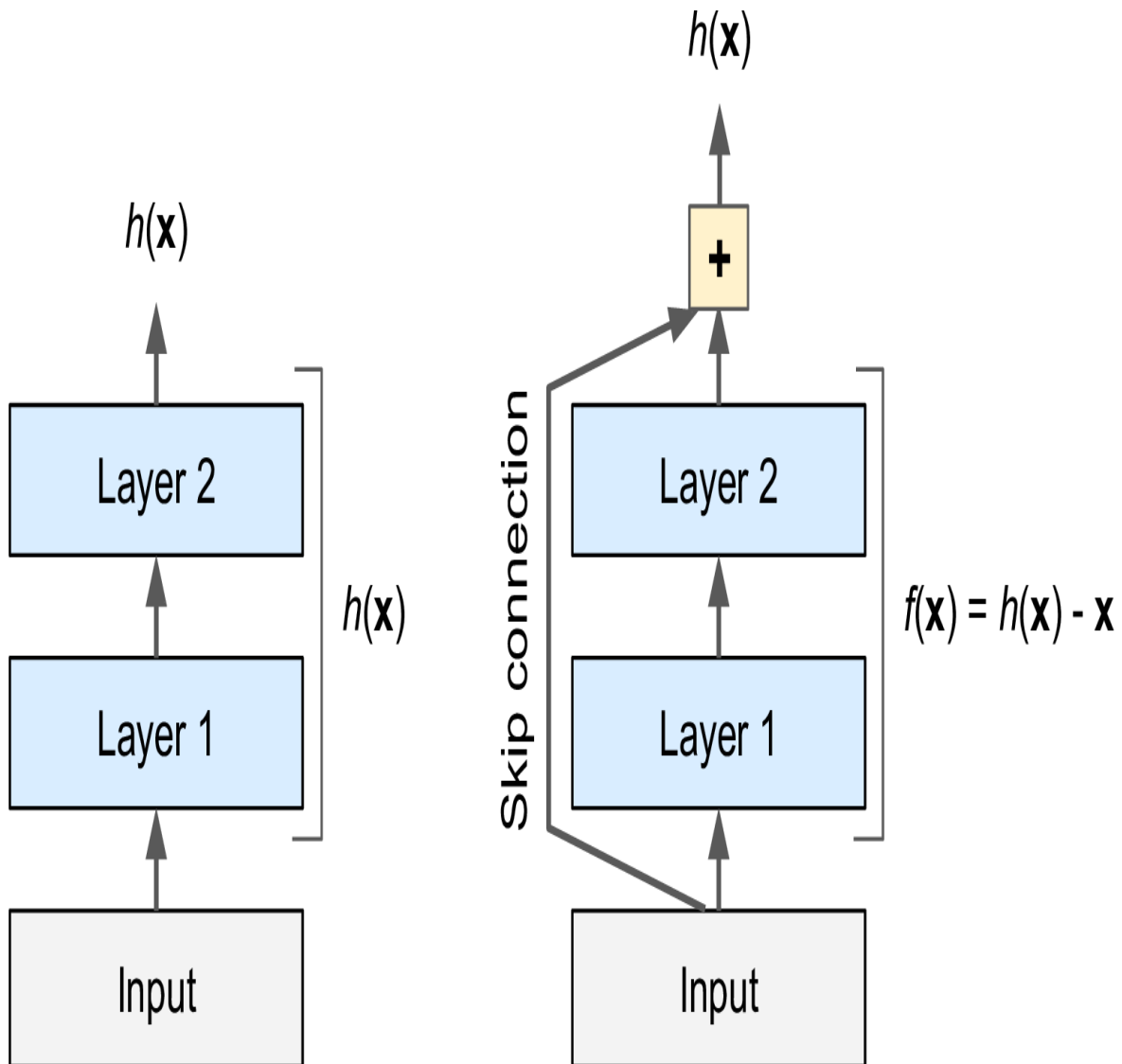


Figure 14-16. Residual learning

When you initialize a regular neural network, its weights are close to zero, so the network just outputs values close to zero. If you add a skip connection, the resulting network just outputs a copy of its inputs; in other words, it initially models the identity function. If the target function is fairly close to the identity function (which is often the case), this will speed up training considerably.

Moreover, if you add many skip connections, the network can start making progress even if several layers have not started learning yet (see [Figure 14-17](#)). Thanks to skip connections, the signal can easily make its way across the whole network. The deep residual network can be seen as a stack of *residual units* (RUs), where each residual unit is a small neural network with a skip connection.

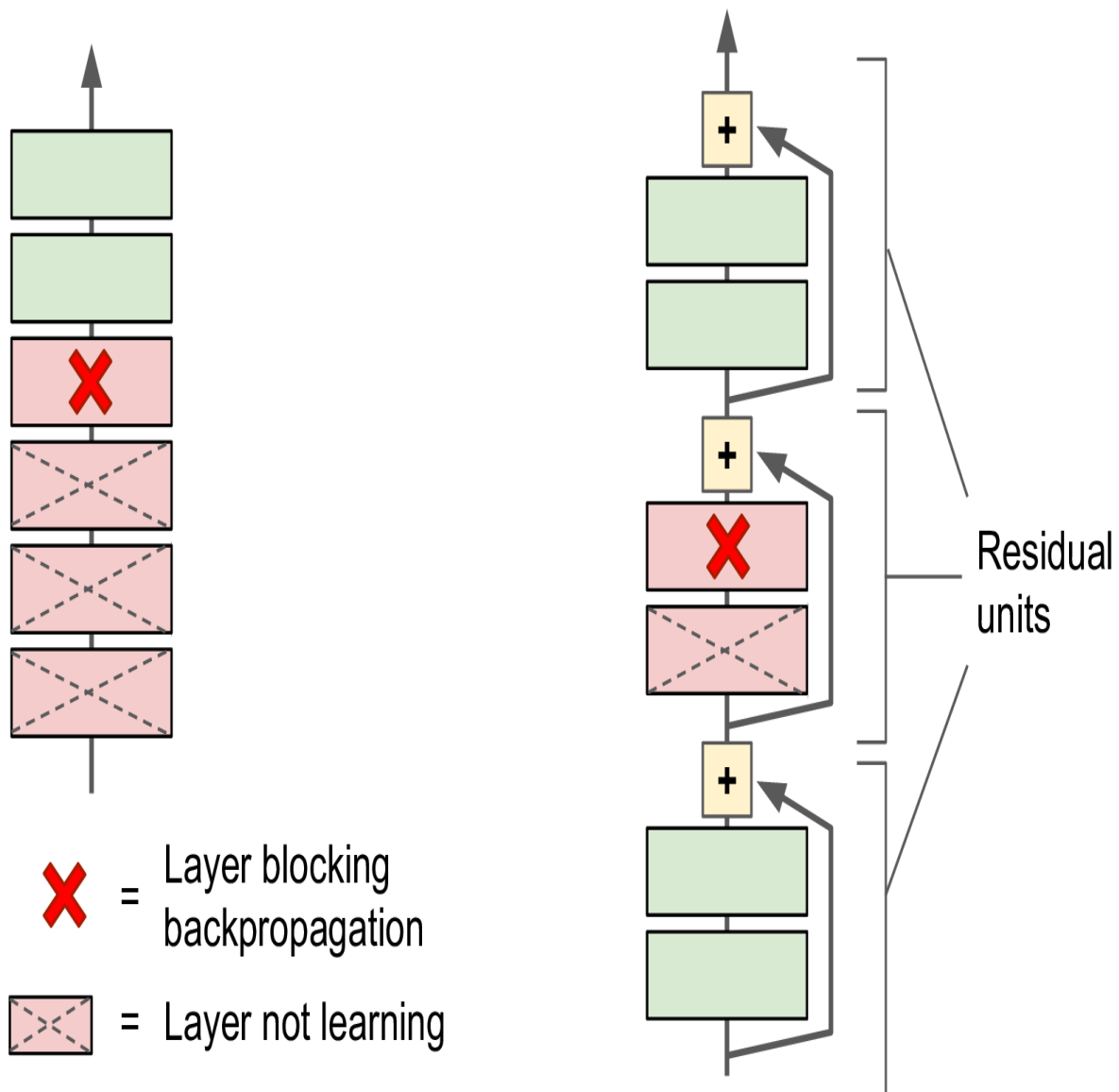


Figure 14-17. Regular deep neural network (left) and deep residual network (right)

Now let's look at ResNet's architecture (see [Figure 14-18](#)). It is surprisingly simple. It starts and ends exactly like GoogLeNet (except without a dropout layer), and in between is just a very deep stack of residual units. Each residual unit is composed of two convolutional layers (and no pooling layer!), with Batch Normalization (BN) and ReLU activation, using 3×3 kernels and preserving spatial dimensions (stride 1, "same" padding).

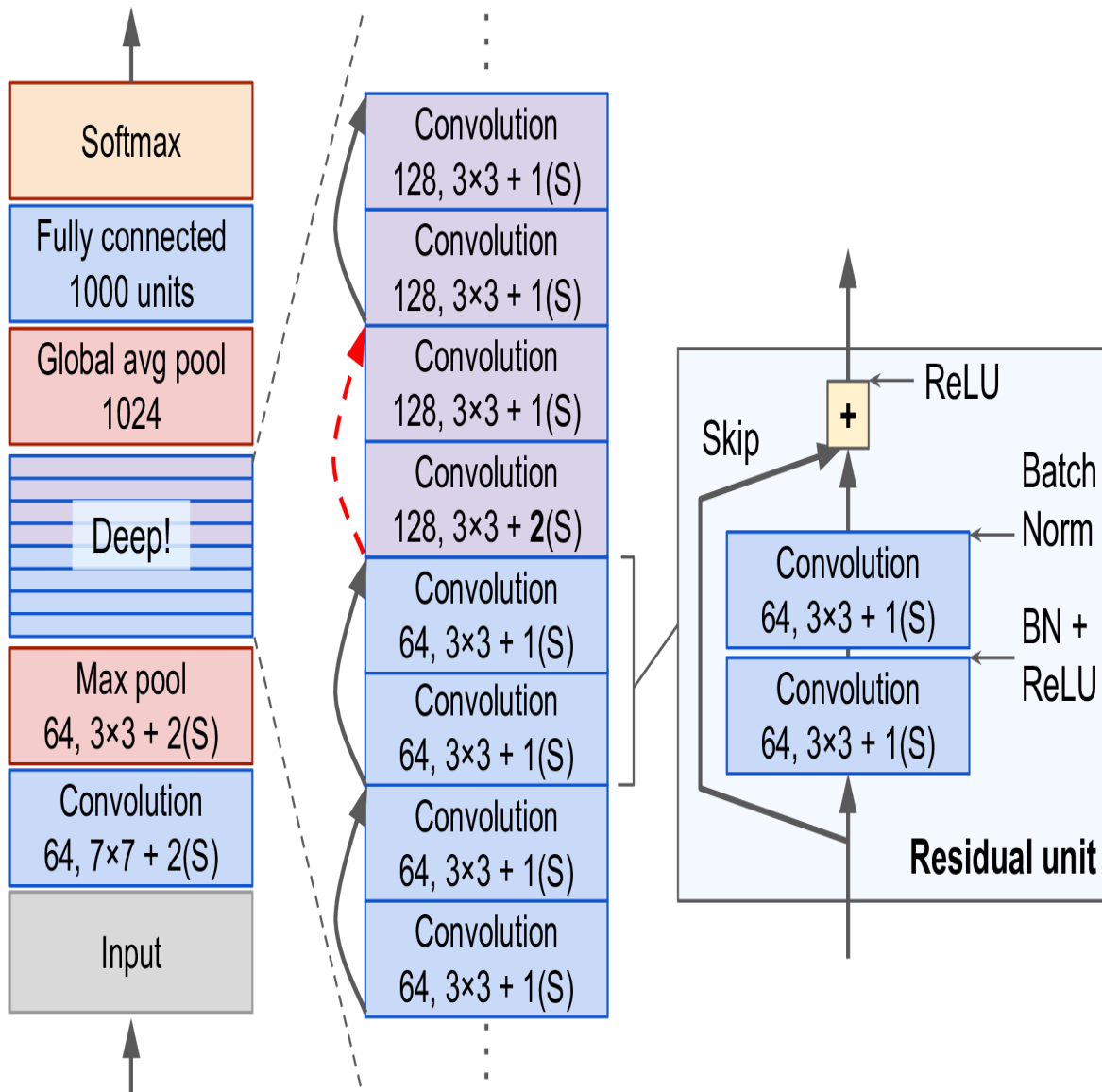


Figure 14-18. ResNet architecture

Note that the number of feature maps is doubled every few residual units, at the same time as their height and width are halved (using a convolutional layer with stride 2). When this happens, the inputs cannot be added directly to the outputs of the residual unit because they don't have the same shape (for example, this problem affects the skip connection represented by the dashed arrow in [Figure 14-18](#)). To solve this problem, the inputs are passed through a 1×1 convolutional layer with stride 2 and the right number of output feature maps (see [Figure 14-19](#)).

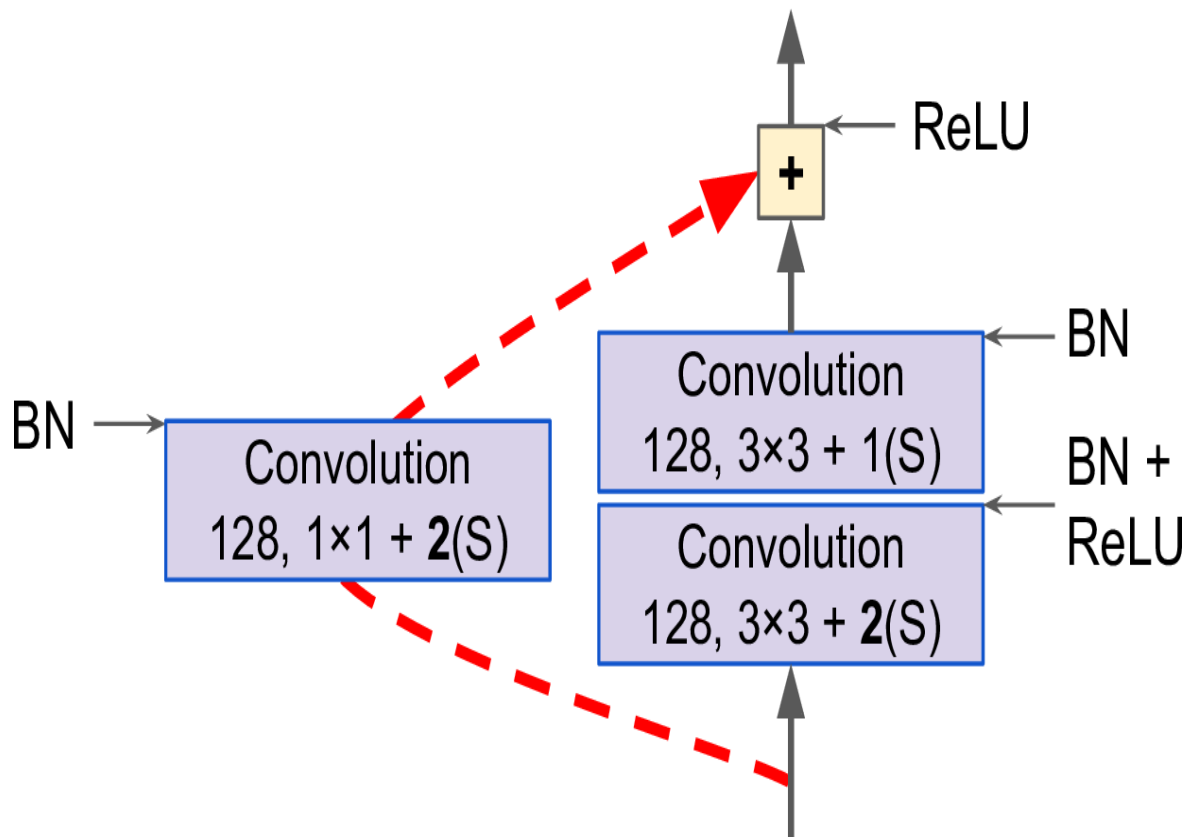


Figure 14-19. Skip connection when changing feature map size and depth

ResNet-34 is the ResNet with 34 layers (only counting the convolutional layers and the fully connected layer)¹⁷ containing 3 residual units that output 64 feature maps, 4 RUs with 128 maps, 6 RUs with 256 maps, and 3 RUs with 512 maps. We will implement this architecture later in this chapter.

ResNets deeper than that, such as ResNet-152, use slightly different residual units. Instead of two 3×3 convolutional layers with, say, 256 feature maps, they use three convolutional layers: first a 1×1 convolutional layer with just 64 feature maps (4 times less), which acts as a bottleneck layer (as discussed already), then a 3×3 layer with 64 feature maps, and finally another 1×1 convolutional layer with 256 feature maps (4 times 64) that restores the original depth. ResNet-152 contains 3 such RUs that output 256 maps, then 8 RUs with 512 maps, a whopping 36 RUs with 1,024 maps, and finally 3 RUs with 2,048 maps.

NOTE

Google's Inception-v4¹⁸ architecture merged the ideas of GoogLeNet and ResNet and achieved a top-five error rate of close to 3% on ImageNet classification.

Xception

Another variant of the GoogLeNet architecture is worth noting: Xception¹⁹ (which stands for *Extreme Inception*) was proposed in 2016 by François Chollet (the author of Keras), and it significantly outperformed Inception-v3 on a huge vision task (350 million images and 17,000 classes). Just like Inception-v4, it merges the ideas of GoogLeNet and ResNet, but it replaces the inception modules with a special type of layer called a *depthwise separable convolution layer* (or *separable convolution layer* for short²⁰). These layers had been used before in some CNN architectures, but they were not as central as in the Xception architecture. While a regular convolutional layer uses filters that try to simultaneously capture spatial patterns (e.g., an oval) and cross-channel

patterns (e.g., mouth + nose + eyes = face), a separable convolutional layer makes the strong assumption that spatial patterns and cross-channel patterns can be modeled separately (see [Figure 14-20](#)). Thus, it is composed of two parts: the first part applies a single spatial filter to each input feature map, then the second part looks exclusively for cross-channel patterns—it is just a regular convolutional layer with 1×1 filters.

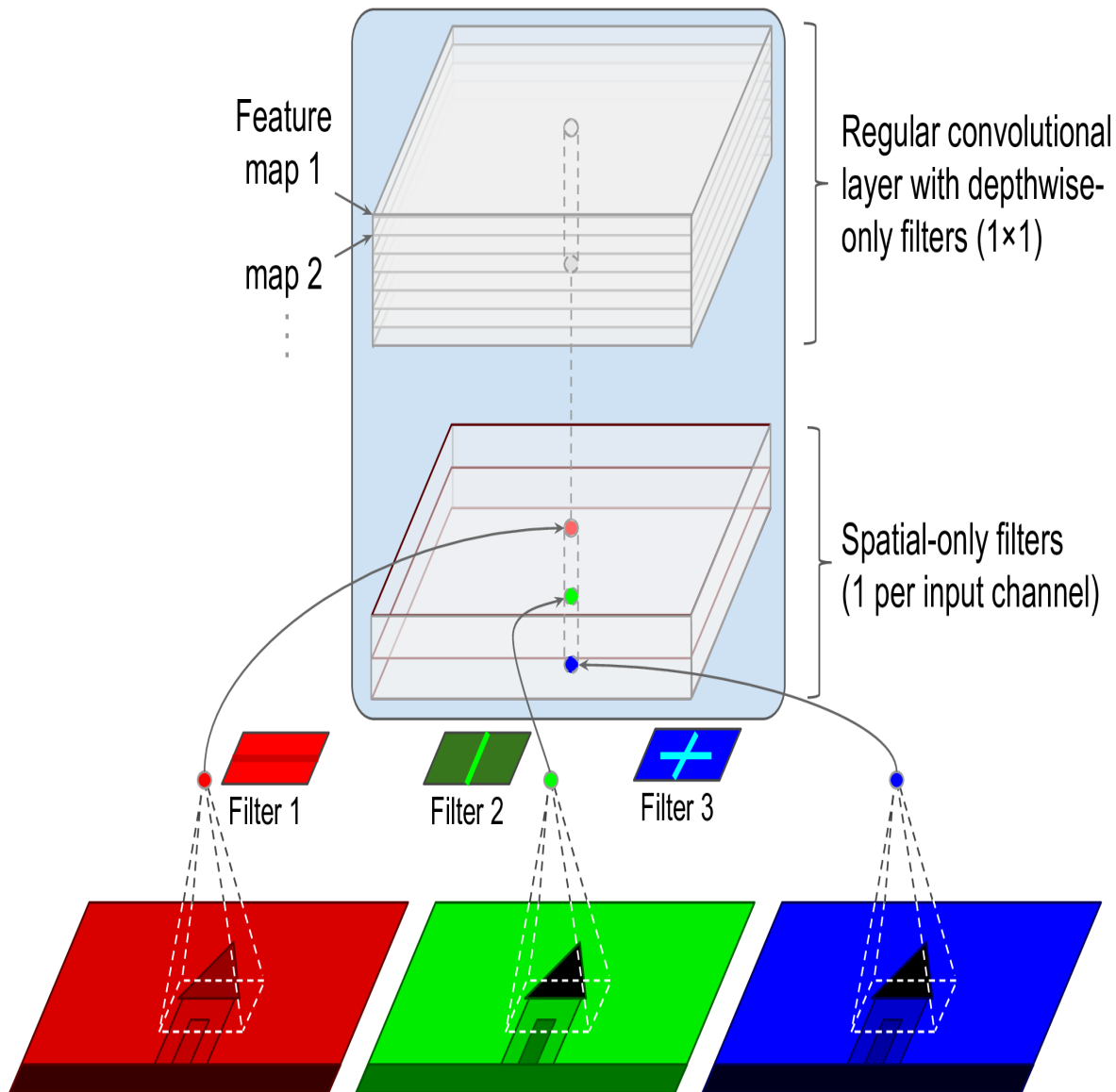


Figure 14-20. Depthwise separable convolutional layer

Since separable convolutional layers only have one spatial filter per input channel, you should avoid using them after layers that have too few channels, such as the input layer (granted, that's what [Figure 14-20](#) represents, but it is just for illustration purposes). For this reason, the Xception architecture starts with 2 regular convolutional layers, but then the rest of the architecture uses only separable convolutions (34 in all), plus a few max pooling layers and the usual final layers (a global average pooling layer and a dense output layer).

You might wonder why Xception is considered a variant of GoogLeNet, since it contains no inception module at all. Well, as we discussed earlier, an inception module contains convolutional layers with 1×1 filters: these look exclusively for cross-channel patterns. However, the convolutional layers that sit on top of them are regular convolutional layers that look both for spatial and cross-channel patterns. So you can think of an inception module as an intermediate between a regular convolutional layer (which considers spatial patterns and cross-channel

patterns jointly) and a separable convolutional layer (which considers them separately). In practice, it seems that separable convolutional layers often perform better.

TIP

Separable convolutional layers use fewer parameters, less memory, and fewer computations than regular convolutional layers, and they often perform better. You should consider using them by default, except after layers with few channels (such as the input channel). In Keras, just use `SeparableConv2D` instead of `Conv2D`: it's a drop-in replacement. Keras also offers a `DepthwiseConv2D` layer which implements the first part of a depthwise separable convolutional layer (i.e., applying one spatial filter per input feature map).

SENet

The winning architecture in the ILSVRC 2017 challenge was the **Squeeze-and-Excitation Network (SENet)**.²¹ This architecture extends existing architectures such as inception networks and ResNets, and boosts their performance. This allowed SENet to win the competition with an astonishing 2.25% top-five error rate! The extended versions of inception networks and ResNets are called *SE-Inception* and *SE-ResNet*, respectively. The boost comes from the fact that a SENet adds a small neural network, called an *SE block*, to every inception module or residual unit in the original architecture, as shown in **Figure 14-21**.

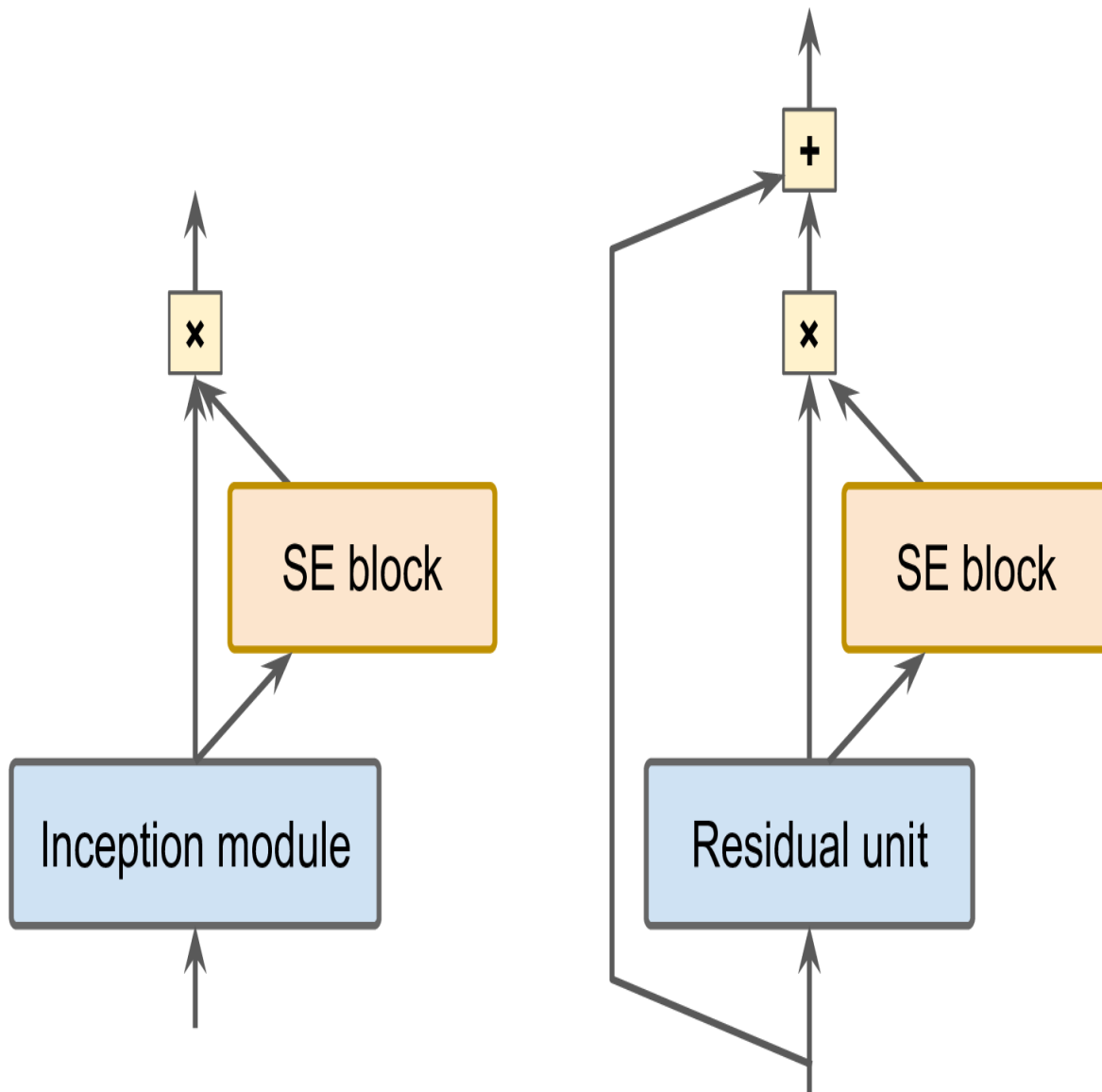


Figure 14-21. SE-Inception module (left) and SE-ResNet unit (right)

An SE block analyzes the output of the unit it is attached to, focusing exclusively on the depth dimension (it does not look for any spatial pattern), and it learns which features are usually most active together. It then uses this information to recalibrate the feature maps, as shown in [Figure 14-22](#). For example, an SE block may learn that mouths, noses, and eyes usually appear together in pictures: if you see a mouth and a nose, you should expect to see eyes as well. So if the block sees a strong activation in the mouth and nose feature maps, but only mild activation in the eye feature map, it will boost the eye feature map (more accurately, it will reduce irrelevant feature maps). If the eyes were somewhat confused with something else, this feature map recalibration will help resolve the ambiguity.

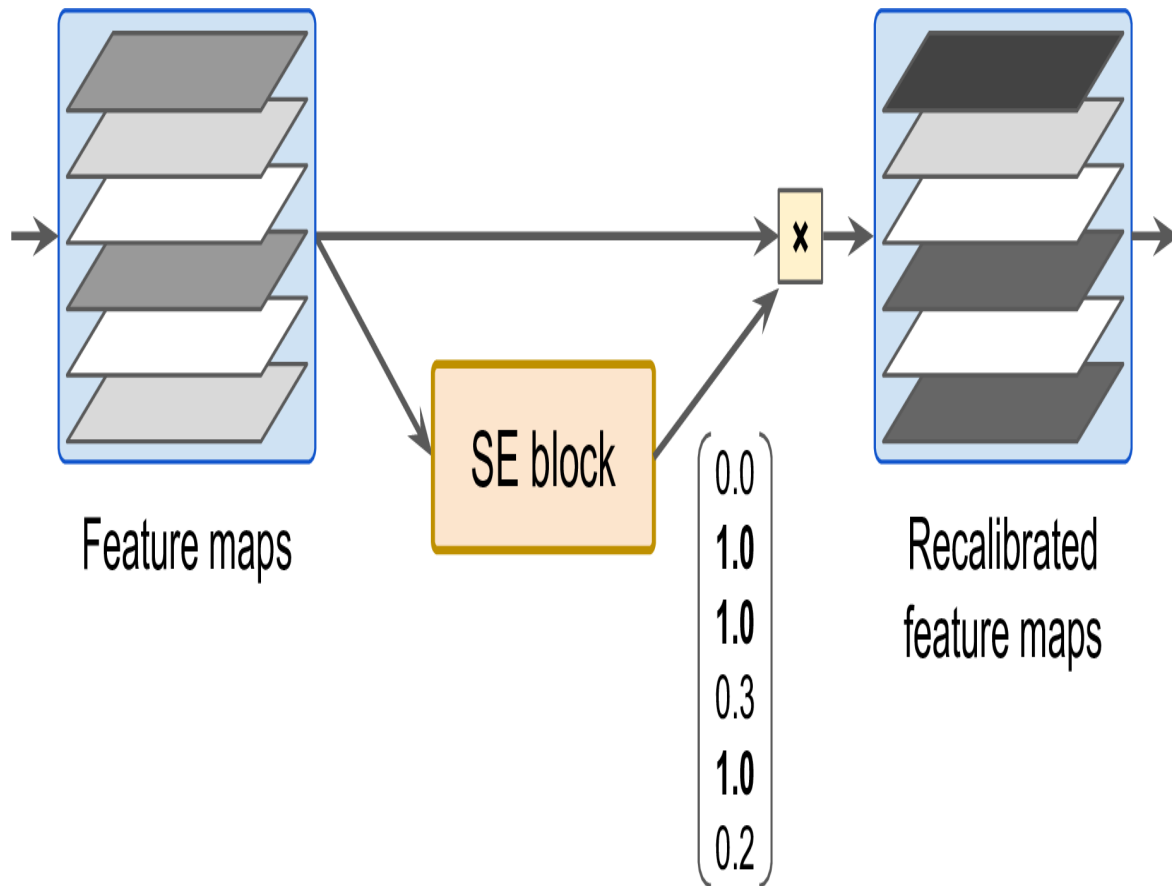


Figure 14-22. An SE block performs feature map recalibration

An SE block is composed of just three layers: a global average pooling layer, a hidden dense layer using the ReLU activation function, and a dense output layer using the sigmoid activation function (see [Figure 14-23](#)).

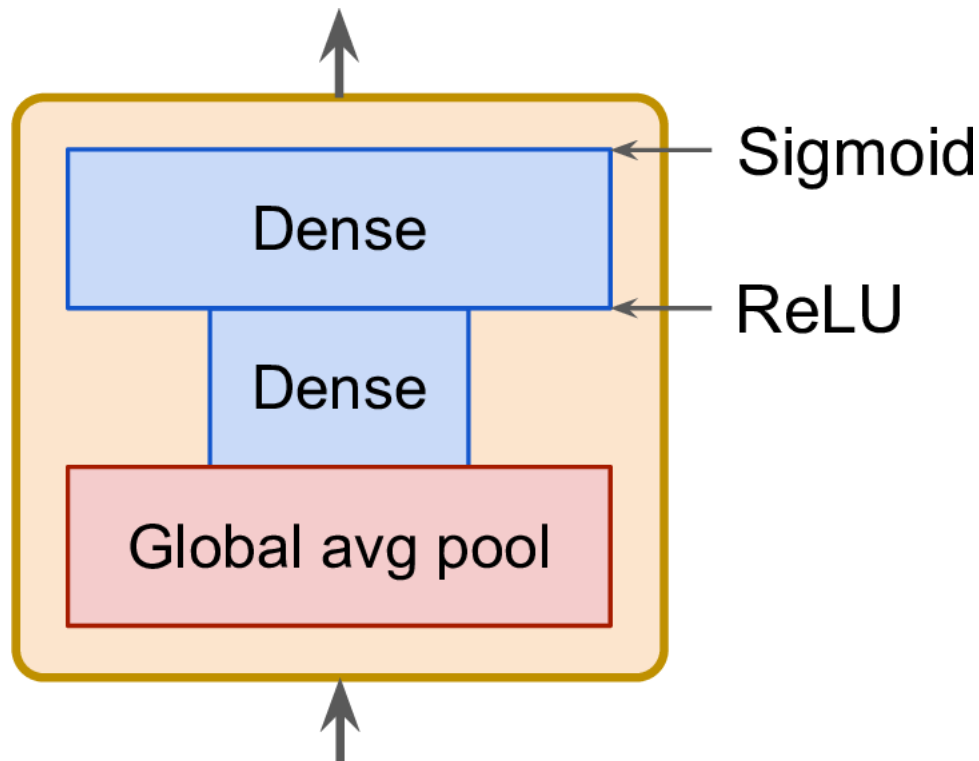


Figure 14-23. SE block architecture

As earlier, the global average pooling layer computes the mean activation for each feature map: for example, if its input contains 256 feature maps, it will output 256 numbers representing the overall level of response for each filter. The next layer is where the “squeeze” happens: this layer has significantly fewer than 256 neurons—typically 16 times fewer than the number of feature maps (e.g., 16 neurons)—so the 256 numbers get compressed into a small vector (e.g., 16 dimensions). This is a low-dimensional vector representation (i.e., an embedding) of the distribution of feature responses. This bottleneck step forces the SE block to learn a general representation of the feature combinations (we will see this principle in action again when we discuss autoencoders in [Chapter 17](#)). Finally, the output layer takes the embedding and outputs a recalibration vector containing one number per feature map (e.g., 256), each between 0 and 1. The feature maps are then multiplied by this recalibration vector, so irrelevant features (with a low recalibration score) get scaled down while relevant features (with a recalibration score close to 1) are left alone.

Other Noteworthy Architectures

There are many other CNN architectures to explore. Here’s a brief overview of some of the most noteworthy:

*ResNeXt*²²

ResNeXt improves the residual units in ResNet. Whereas the residual units in the best ResNet models just contain 3 convolution layers each, the ResNeXt residual units are composed of many parallel stacks (e.g., 32 stacks), with 3 convolutional layers each. However, the first two layers in each stack only use a few filters (e.g., just 4), so the overall number of parameters remains the same as in ResNet. Then the outputs of all stacks are added together, and the result is passed to the next residual unit (along with the skip connection).

*DenseNet*²³

A DenseNet is composed of several dense blocks, each made up of a few densely connected convolutional layers. What does “densely connected” mean? Well, the output of each layer is fed as input to every layer after it within the same block. For example, layer 4 in a block takes as input the depthwise concatenation of the

outputs of layers 1, 2 and 3 in that block. Dense blocks are separated by a few transition layers. This architecture achieved excellent accuracy while using comparatively few parameters.

MobileNet²⁴

MobileNets are streamlined models designed to be lightweight and fast, making them popular in mobile and web applications. They are based on depthwise separable convolutional layers, like Xception. The authors proposed several variants, trading a bit of accuracy for faster and smaller models.

CSPNet²⁵

Cross Stage Partial Network (CSPNet) is similar to DenseNet, but part of each dense block's input is concatenated directly to that block's output, without going through the block.

EfficientNet²⁶

EfficientNet is arguably the most important model in this list. The authors proposed a method to scale any CNN efficiently, by jointly increasing the depth (number of layers), width (number of filters per layer), and resolution (the size of the input image), in a principled way (explained below). This is called *compound scaling*. They used neural architecture search to find a good architecture for a scaled down version of ImageNet (with smaller and fewer images), then they used compound scaling to create larger and larger versions of this architecture. When EfficientNet models came out, they vastly outperformed all existing models, across all compute budgets, and they remain among the best models out there today.

Understanding EfficientNet's compound scaling method is helpful to gain a deeper understanding of CNNs, especially if you ever need to scale a CNN architecture. It is based on a logarithmic measure of the compute budget noted ϕ : if your compute budget doubles, then ϕ increases by 1. In other words, the number of floating-point operations available for training is proportional to 2^ϕ . Your CNN architecture's depth, width, and resolution should scale as α^ϕ , β^ϕ , and γ^ϕ , respectively. The factors α , β , and γ must be greater than 1, and $\alpha + \beta^2 + \gamma^2$ should be close to 2. The optimal values for these factors depend on the CNN's architecture. To find the optimal values for the EfficientNet architecture, the authors started with a small baseline model (EfficientNet-B0), they fixed $\phi = 1$, and they simply ran a grid search: they found $\alpha = 1.2$, $\beta = 1.1$, and $\gamma = 1.1$. They then used these factors to create several larger architectures, named EfficientNet-B1 to EfficientNet-B7, for increasing values of ϕ .

Choosing The Right CNN Architecture

OK, with so many CNN architectures, how do you choose which one is best for your project? Well, it depends on what matters most to you: accuracy? Model size (e.g., for deployment to a mobile device)? Inference speed on CPU? On GPU? Table 14-3 lists the best pretrained models currently available in Keras (we will see how to use them later in this chapter), sorted by model size. You can find the full list at <https://keras.io/api/applications/>. For each model, the table shows the Keras class name to use (in the `tf.keras.applications` package), the model's size in MB, the top-1 and top-5 validation accuracy on the ImageNet dataset, the number of parameters (millions), and the inference time on CPU and GPU in ms, using batches of 32 images on reasonably powerful hardware.²⁷ For each column, the best value is highlighted. As you can see, larger models are generally more accurate, but not always. For example, EfficientNetB2 outperforms InceptionV3 both in size and accuracy. I only kept InceptionV3 in the list because it is almost twice as fast as EfficientNetB2 on a CPU. Similarly, InceptionResNetV2 is fast on a CPU, and ResNet50V2 and ResNet101V2 are blazingly fast on a GPU.

Table 14 - 3. Performance of models

Class Name	Size (MB)	Top-1 Acc	Top-5 Acc	Params	CPU (ms)	GPU (ms)
MobileNetV2	14	71.3%	90.1%	3.5M	25.9	3.8
MobileNet	16	70.4%	89.5%	4.3M	22.6	3.4
NASNetMobile	23	74.4%	91.9%	5.3M	27.0	6.7
EfficientNetB0	29	77.1%	93.3%	5.3M	46.0	4.9
EfficientNetB1	31	79.1%	94.4%	7.9M	60.2	5.6
EfficientNetB2	36	80.1%	94.9%	9.2M	80.8	6.5
EfficientNetB3	48	81.6%	95.7%	12.3M	140.0	8.8

EfficientNetB4	75	82.9%	96.4%	19.5M	308.3	15.1
InceptionV3	92	77.9%	93.7%	23.9M	42.2	6.9
ResNet50V2	98	76.0%	93.0%	25.6M	45.6	4.4
EfficientNetB5	118	83.6%	96.7%	30.6M	579.2	25.3
EfficientNetB6	166	84.0%	96.8%	43.3M	958.1	40.4
ResNet101V2	171	77.2%	93.8%	44.7M	72.7	5.4
InceptionResNetV2	215	80.3%	95.3%	55.9M	130.2	10.0
EfficientNetB7	256	84.3%	97.0%	66.7M	1578.9	61.6

I hope you enjoyed this deep dive into the main CNN architectures! Now let's see how to implement one of them using Keras.

Implementing a ResNet-34 CNN Using Keras

Most CNN architectures described so far can be implemented pretty naturally using Keras (although generally you would load a pretrained network instead, as we will see). To illustrate the process, let's implement a ResNet-34 from scratch using Keras. First, let's create a `ResidualUnit` layer:

```
DefaultConv2D = partial(tf.keras.layers.Conv2D, kernel_size=3, strides=1,
                        padding="same", kernel_initializer="he_normal",
                        use_bias=False)

class ResidualUnit(tf.keras.layers.Layer):
    def __init__(self, filters, strides=1, activation="relu", **kwargs):
        super().__init__(**kwargs)
        self.activation = tf.keras.activations.get(activation)
        self.main_layers = [
            DefaultConv2D(filters, strides=strides),
            tf.keras.layers.BatchNormalization(),
            self.activation,
            DefaultConv2D(filters),
            tf.keras.layers.BatchNormalization()
        ]
        self.skip_layers = []
        if strides > 1:
            self.skip_layers = [
                DefaultConv2D(filters, kernel_size=1, strides=strides),
                tf.keras.layers.BatchNormalization()
            ]

    def call(self, inputs):
        Z = inputs
        for layer in self.main_layers:
            Z = layer(Z)
        skip_Z = inputs
        for layer in self.skip_layers:
            skip_Z = layer(skip_Z)
        return self.activation(Z + skip_Z)
```

As you can see, this code matches [Figure 14-19](#) pretty closely. In the constructor, we create all the layers we will need: the main layers are the ones on the right side of the diagram, and the skip layers are the ones on the left (only needed if the stride is greater than 1). Then in the `call()` method, we make the inputs go through the main layers and the skip layers (if any), then we add both outputs and apply the activation function.

Next, we can build ResNet-34 using a `Sequential` model, since it's really just a long sequence of layers—we can treat each residual unit as a single layer now that we have the `ResidualUnit` class. The code closely matches [Figure 14-18](#):

```
model = tf.keras.Sequential([
    DefaultConv2D(64, kernel_size=7, strides=2, input_shape=[224, 224, 3]),
    tf.keras.layers.BatchNormalization(),
    tf.keras.layers.Activation("relu"),
    tf.keras.layers.MaxPool2D(pool_size=3, strides=2, padding="same"),
])
prev_filters = 64
for filters in [64] * 3 + [128] * 4 + [256] * 6 + [512] * 3:
    strides = 1 if filters == prev_filters else 2
    model.add(ResidualUnit(filters, strides=strides))
    prev_filters = filters

model.add(tf.keras.layers.GlobalAvgPool2D())
model.add(tf.keras.layers.Flatten())
model.add(tf.keras.layers.Dense(10, activation="softmax"))
```

The only tricky part in this code is the loop that adds the `ResidualUnit` layers to the model: as explained earlier, the first 3 RUs have 64 filters, then the next 4 RUs have 128 filters, and so on. At each iteration, we must set the stride to 1 when the number of filters is the same as in the previous RU, or else we set it to 2, then we add the `ResidualUnit`, and finally we update `prev_filters`.

It is amazing that in about 40 lines of code, we can build the model that won the ILSVRC 2015 challenge! This demonstrates both the elegance of the ResNet model and the expressiveness of the Keras API. Implementing the other CNN architectures is a bit longer, but not much harder. However, Keras comes with several of these architectures built in, so why not use them instead?

Using Pretrained Models from Keras

In general, you won't have to implement standard models like GoogLeNet or ResNet manually, since pretrained networks are readily available with a single line of code in the `tf.keras.applications` package. For example, you can load the ResNet-50 model, pretrained on ImageNet, with the following line of code:

```
model = tf.keras.applications.ResNet50(weights="imagenet")
```

That's all! This will create a ResNet-50 model and download weights pretrained on the ImageNet dataset. To use it, you first need to ensure that the images have the right size. A ResNet-50 model expects 224×224 -pixel images (other models may expect other sizes, such as 299×299), so let's use Keras's `Resizing` layer (introduced in [Chapter 13](#)) to resize two sample images (after cropping them to the target aspect ratio):

```
images = load_sample_images()["images"]
images_resized = tf.keras.layers.Resizing(height=224, width=224,
                                          crop_to_aspect_ratio=True)(images)
```

The pretrained models assume that the images are preprocessed in a specific way. In some cases they may expect the inputs to be scaled from 0 to 1, or from -1 to 1 , and so on. Each model provides a `preprocess_input()` function that you can use to preprocess your images. These functions assume that the original pixel values range from 0 to 255, which is the case here.

```
inputs = tf.keras.applications.resnet50.preprocess_input(images_resized)
```

Now we can use the pretrained model to make predictions:


```
>>> Y_proba = model.predict(inputs)
>>> Y_proba.shape
(2, 1000)
```

As usual, the output `Y_proba` is a matrix with one row per image and one column per class (in this case, there are 1,000 classes). If you want to display the top *K* predictions, including the class name and the estimated probability of each predicted class, use the `decode_predictions()` function. For each image, it returns an array containing the top *K* predictions, where each prediction is represented as an array containing the class identifier,²⁸ its name, and the corresponding confidence score:

```
top_K = tf.keras.applications.resnet50.decode_predictions(Y_proba, top=3)
for image_index in range(len(images)):
    print(f"Image #{image_index}")
    for class_id, name, y_proba in top_K[image_index]:
        print(f" {class_id} - {name:12s} {y_proba:.2%}")
```

The output looks like this:

```
Image #0
n03877845 - palace      54.69%
n03781244 - monastery   24.72%
n02825657 - bell_cote   18.55%
Image #1
n04522168 - vase        32.66%
n11939491 - daisy       17.81%
n03530642 - honeycomb   12.06%
```

The correct classes are *palace* and *dahlia*, so the model is correct for the first image but wrong for the second. However, that's because *dahlia* is not part of the 1,000 ImageNet classes. With that in mind, *vase* is a reasonable guess (perhaps the flower is in a vase?), and daisy is not a bad choice either, since dalias and daisies are both from the same Compositae family.

As you can see, it is very easy to create a pretty good image classifier using a pretrained model. As we saw in [Table 14-3](#), many other vision models are available in `tf.keras.applications`, from lightweight and fast models to large and accurate ones.

But what if you want to use an image classifier for classes of images that are not part of ImageNet? In that case, you may still benefit from the pretrained models to perform transfer learning.

Pretrained Models for Transfer Learning

If you want to build an image classifier but you do not have enough data to train it from scratch, then it is often a good idea to reuse the lower layers of a pretrained model, as we discussed in [Chapter 11](#). For example, let's train a model to classify pictures of flowers, reusing a pretrained Xception model. First, let's load the dataset using TensorFlow Datasets (introduced in [Chapter 13](#)):

```
import tensorflow_datasets as tfds

dataset, info = tfds.load("tf_flowers", as_supervised=True, with_info=True)
dataset_size = info.splits["train"].num_examples # 3670
class_names = info.features["label"].names # ["dandelion", "daisy", ...]
n_classes = info.features["label"].num_classes # 5
```

Note that you can get information about the dataset by setting `with_info=True`. Here, we get the dataset size and the names of the classes. Unfortunately, there is only a "train" dataset, no test set or validation set, so we need to split the training set. Let's call `tfds.load()` again, but this time taking the first 10% of the dataset for testing, the next 15% for validation, and the remaining 75% for training:

```
test_set_raw, valid_set_raw, train_set_raw = tfds.load(
    "tf_flowers",
    split=["train[:10%]", "train[10%:25%]", "train[25%:]"],
    as_supervised=True)
```

All three datasets contain individual images. We need to batch them, but for this we must first ensure they all have the same size, or else batching will fail. We can use a `Resizing` layer for this. We must also call the `tf.keras.applications.xception.preprocess_input()` function to preprocess the images appropriately for the Xception model. Lastly, let's also shuffle the training set and use prefetching:

```
batch_size = 32
preprocess = tf.keras.Sequential([
    tf.keras.layers.Resizing(height=224, width=224, crop_to_aspect_ratio=True),
    tf.keras.layers.Lambda(tf.keras.applications.xception.preprocess_input)
])
train_set = train_set_raw.map(lambda X, y: (preprocess(X), y))
train_set = train_set.shuffle(1000, seed=42).batch(batch_size).prefetch(1)
valid_set = valid_set_raw.map(lambda X, y: (preprocess(X), y)).batch(batch_size)
test_set = test_set_raw.map(lambda X, y: (preprocess(X), y)).batch(batch_size)
```

Now each batch contains 32 images, all of them 224×224 pixels, with pixel values ranging from -1 to 1 . Perfect!

Since the dataset is not very large, a bit of data augmentation will certainly help. Let's create a data augmentation model that we will embed in our final model: during training, it will randomly flip the images horizontally, rotate them a little bit, and tweak the contrast.

```
data_augmentation = tf.keras.Sequential([
    tf.keras.layers.RandomFlip(mode="horizontal", seed=42),
    tf.keras.layers.RandomRotation(factor=0.05, seed=42),
    tf.keras.layers.RandomContrast(factor=0.2, seed=42)
])
```

TIP

The `tf.keras.preprocessing.image.ImageDataGenerator` class makes it easy to load images from disk and augment them in various ways: you can shift each image, rotate it, rescale it, flip it horizontally or vertically, shear it, or apply any transformation function you want to it. This is very convenient for simple projects. However, a `tf.data` pipeline is not much more complicated, and it's generally faster. Moreover, if you have a GPU and you include the preprocessing or data augmentation layers inside your model, they will benefit from GPU acceleration during training.

Next let's load an Xception model, pretrained on ImageNet. We exclude the top of the network by setting `include_top=False`. This excludes the global average pooling layer and the dense output layer. We then add our own global average pooling layer (feeding it the output of the base model), followed by a dense output layer with one unit per class, using the softmax activation function. Finally, we wrap all this in a Keras `Model`:

```
base_model = tf.keras.applications.xception.Xception(weights="imagenet",
    include_top=False)
avg = tf.keras.layers.GlobalAveragePooling2D()(base_model.output)
output = tf.keras.layers.Dense(n_classes, activation="softmax")(avg)
model = tf.keras.Model(inputs=base_model.input, outputs=output)
```

As explained in [Chapter 11](#), it's usually a good idea to freeze the weights of the pretrained layers, at least at the beginning of training:

```
for layer in base_model.layers:
    layer.trainable = False
```



```
model.compile(loss=["sparse_categorical_crossentropy", "mse"],
              loss_weights=[0.8, 0.2], # depends on what you care most about
              optimizer=optimizer, metrics=["accuracy"])
```

But now we have a problem: the flowers dataset does not have bounding boxes around the flowers. So, we need to add them ourselves. This is often one of the hardest and most costly parts of a Machine Learning project: getting the labels. It's a good idea to spend time looking for the right tools. To annotate images with bounding boxes, you may want to use an open source image labeling tool like VGG Image Annotator, LabelImg, OpenLabeler, or ImgLab, or perhaps a commercial tool like LabelBox or Supervisely. You may also want to consider crowdsourcing platforms such as Amazon Mechanical Turk if you have a very large number of images to annotate. However, it is quite a lot of work to set up a crowdsourcing platform, prepare the form to be sent to the workers, supervise them, and ensure that the quality of the bounding boxes they produce is good, so make sure it is worth the effort. Adriana Kovashka et al. wrote a very practical [paper](#)²⁹ about crowdsourcing in computer vision. I recommend you check it out, even if you do not plan to use crowdsourcing. If there are just a few hundred or a even a couple thousand images to label, and you don't plan to do this frequently, it may be preferable to do it yourself: with the right tools, it will only take a few days, and you'll also gain a better understanding of your dataset and task.

Now let's suppose you've obtained the bounding boxes for every image in the flowers dataset (for now we will assume there is a single bounding box per image). You then need to create a dataset whose items will be batches of preprocessed images along with their class labels and their bounding boxes. Each item should be a tuple of the form `(images, (class_labels, bounding_boxes))`. Then you are ready to train your model!

TIP

The bounding boxes should be normalized so that the horizontal and vertical coordinates, as well as the height and width, all range from 0 to 1. Also, it is common to predict the square root of the height and width rather than the height and width directly: this way, a 10-pixel error for a large bounding box will not be penalized as much as a 10-pixel error for a small bounding box.

The MSE often works fairly well as a cost function to train the model, but it is not a great metric to evaluate how well the model can predict bounding boxes. The most common metric for this is the *Intersection over Union* (IoU): the area of overlap between the predicted bounding box and the target bounding box, divided by the area of their union (see [Figure 14-24](#)). In Keras, it is implemented by the `tf.keras.metrics.MeanIoU` class.

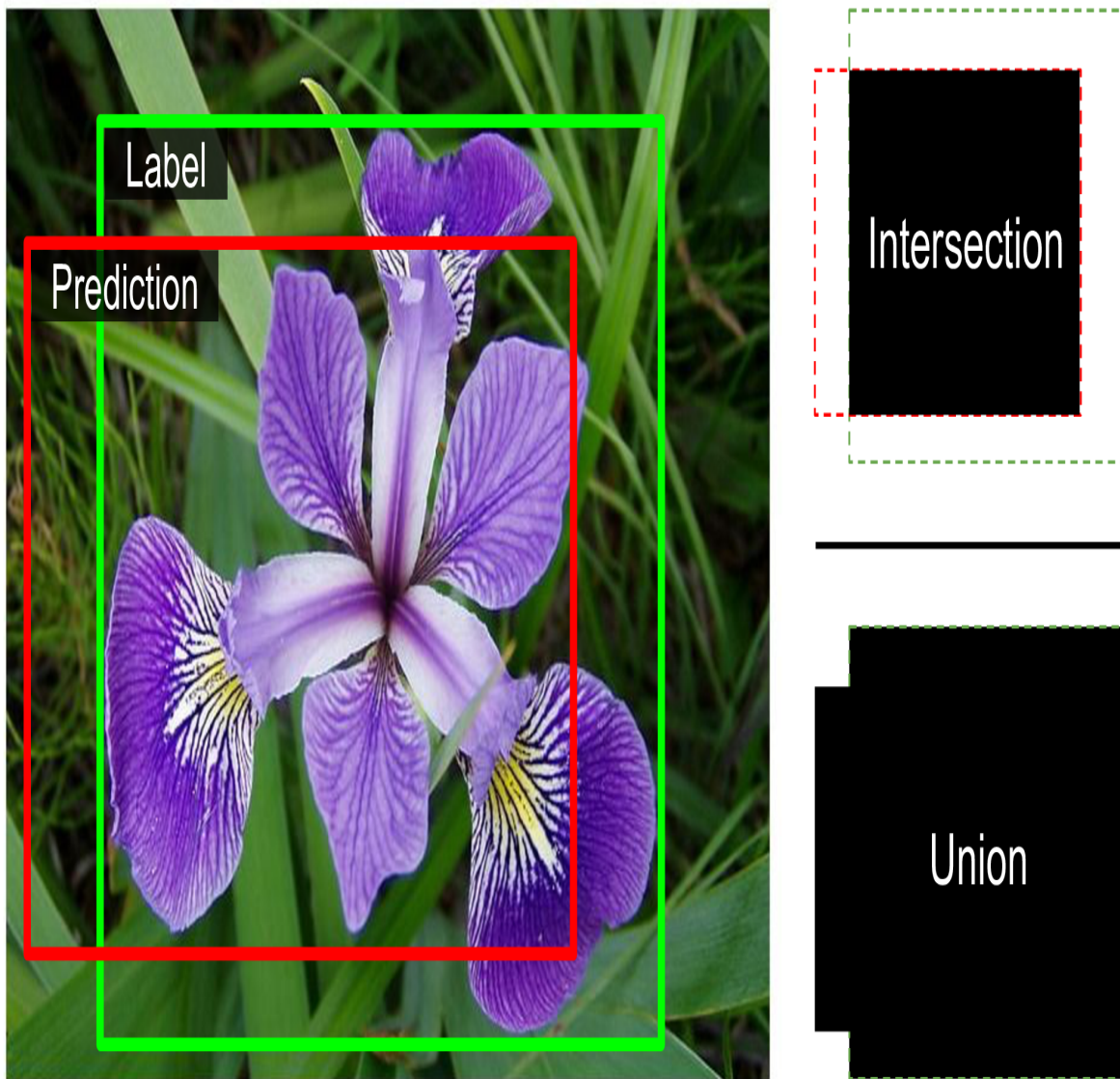


Figure 14-24. Intersection over Union (IoU) metric for bounding boxes

Classifying and localizing a single object is nice, but what if the images contain multiple objects (as is often the case in the flowers dataset)?

Object Detection

The task of classifying and localizing multiple objects in an image is called *object detection*. Until a few years ago, a common approach was to take a CNN that was trained to classify and locate a single object roughly centered in the image, then slide this CNN across the image, and make predictions at each step. The CNN was generally trained to predict not only class probabilities and a bounding box, but also an *objectness score*: this is the estimated probability that the image does indeed contain an object centered near the middle of the image. This is a binary classification output: it can be produced by a dense output layer with a single unit, using the sigmoid activation function and trained using the binary cross-entropy loss.

NOTE

Instead of an objectness score, a “no-object” class was sometimes added, but in general this did not work as well: the questions “is an object present?” and “what type of object is it?” are best answered separately.

This sliding-CNN approach is illustrated in [Figure 14-25](#). In this example, the image was chopped into a 5×7 grid, and we see a CNN—the thick black rectangle—sliding across all 3×3 regions, and making predictions at each step.

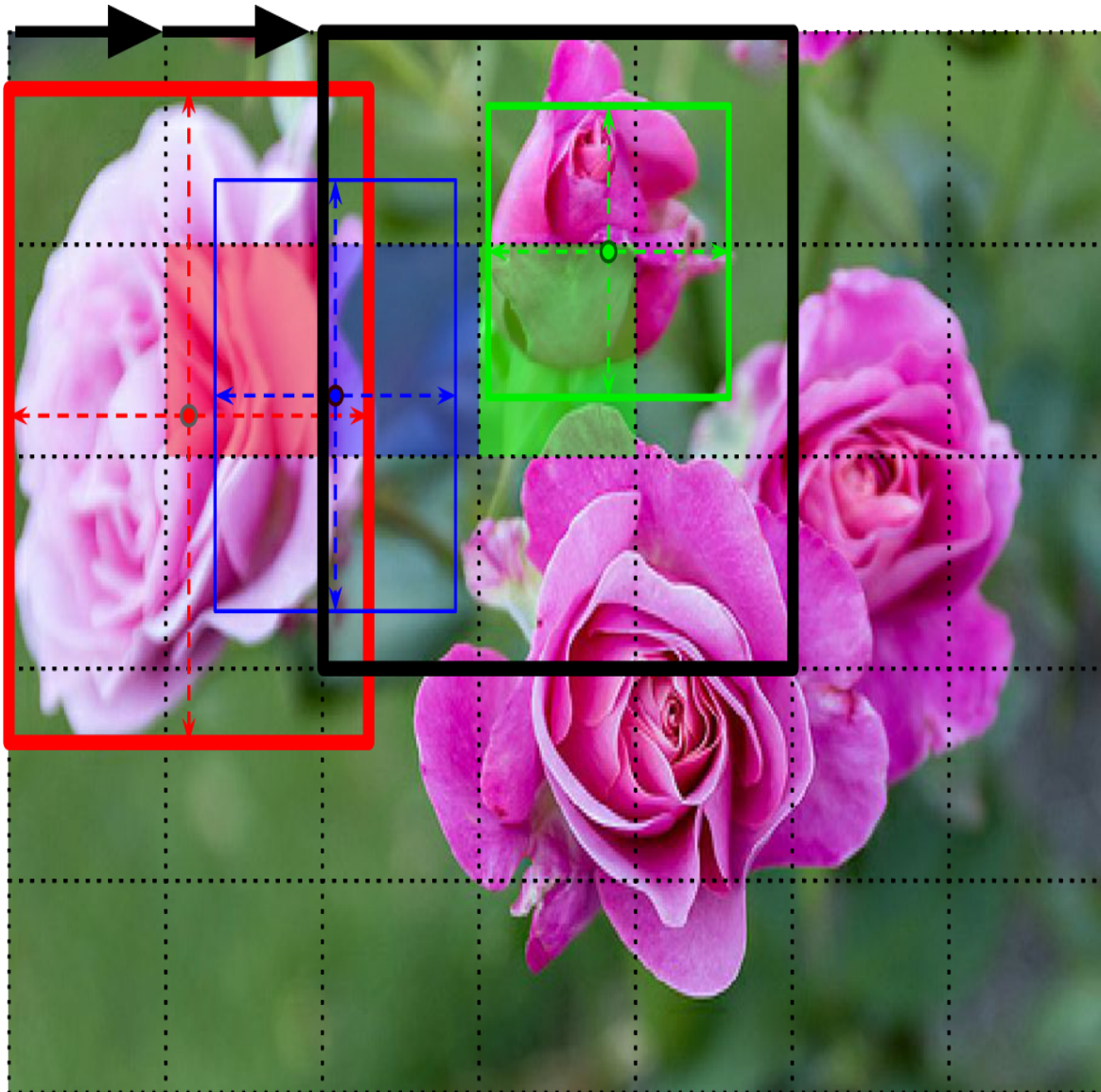


Figure 14-25. Detecting multiple objects by sliding a CNN across the image

In this figure, the CNN has already made predictions for three of these 3×3 regions:

- When looking at the top-left 3×3 region (centered on the red-shaded grid cell located in the second row and second column), it detected the leftmost rose. Notice that the predicted bounding box exceeds the boundary of this 3×3 region. That’s absolutely fine: even though the CNN could not see the bottom part of the rose, it was able to make a reasonable guess as to where it might be. It also predicted class probabilities, giving a high probability to the “rose” class. Lastly, it predicted a fairly high objectness score, since the center of the

bounding box lies within the central grid cell (in this figure, the objectness score is represented by the thickness of the bounding box).

- Then when looking at the next 3×3 region, one grid cell to the right (centered on the shaded blue square), it did not detect any flower centered in that region, so it predicted a very low objectness score, and therefore the predicted bounding box and class probabilities can safely be ignored. You can see that the predicted bounding box was no good anyway.
- Then, once again, it looked at the next 3×3 region, one grid cell to the right (centered on the shaded green cell), and this time it detected the rose at the top, although not perfectly since this rose is not well centered within this region, so the predicted objectness score was not very high.

You can imagine how sliding the CNN across the whole image would give you a total of 15 predicted bounding boxes, organized in a 3×5 grid, with each bounding box accompanied by its estimated class probabilities and objectness score. Since objects can have varying sizes, you may then want to slide the CNN again across larger 4×4 regions as well, to get even more bounding boxes.

This technique is fairly straightforward, but as you can see it will often detect the same object multiple times, at slightly different positions. Some post-processing is needed to get rid of all the unnecessary bounding boxes. A common approach for this is called *non-max suppression*. Here's how it works:

1. First, get rid of all the bounding boxes for which the objectness score is below some threshold: since the CNN believes there's no object at that location, the bounding box is useless.
2. Find the remaining bounding box with the highest objectness score, and get rid of all the other remaining bounding boxes that overlap a lot with it (e.g., with an IoU greater than 60%). For example, in [Figure 14-25](#), the bounding box with the max objectness score is the thick bounding box over the leftmost rose. The other bounding box that touches this same rose overlaps a lot with the max bounding box, so we will get rid of it (although in this example it would already have been removed in the previous step).
3. Repeat step two until there are no more bounding boxes to get rid of.

This simple approach to object detection works pretty well, but it requires running the CNN many times (15 times in this example), so it is quite slow. Fortunately, there is a much faster way to slide a CNN across an image: using a *fully convolutional network* (FCN).

Fully Convolutional Networks

The idea of FCNs was first introduced in a [2015 paper³⁰](#) by Jonathan Long et al., for semantic segmentation (the task of classifying every pixel in an image according to the class of the object it belongs to). The authors pointed out that you could replace the dense layers at the top of a CNN, with convolutional layers. To understand this, let's look at an example: suppose a dense layer with 200 neurons sits on top of a convolutional layer that outputs 100 feature maps, each of size 7×7 (this is the feature map size, not the kernel size). Each neuron will compute a weighted sum of all $100 \times 7 \times 7$ activations from the convolutional layer (plus a bias term). Now let's see what happens if we replace the dense layer with a convolutional layer using 200 filters, each of size 7×7 , and with "valid" padding. This layer will output 200 feature maps, each 1×1 (since the kernel is exactly the size of the input feature maps and we are using "valid" padding). In other words, it will output 200 numbers, just like the dense layer did; and if you look closely at the computations performed by a convolutional layer, you will notice that these numbers will be precisely the same as those the dense layer produced. The only difference is that the dense layer's output was a tensor of shape `[batch size, 200]`, while the convolutional layer will output a tensor of shape `[batch size, 1, 1, 200]`.

TIP

To convert a dense layer to a convolutional layer, the number of filters in the convolutional layer must be equal to the number of units in the dense layer, the filter size must be equal to the size of the input feature maps, and you must use “valid” padding. The stride may be set to 1 or more, as we will see shortly.

Why is this important? Well, while a dense layer expects a specific input size (since it has one weight per input feature), a convolutional layer will happily process images of any size³¹ (however, it does expect its inputs to have a specific number of channels, since each kernel contains a different set of weights for each input channel). Since an FCN contains only convolutional layers (and pooling layers, which have the same property), it can be trained and executed on images of any size!

For example, suppose we’d already trained a CNN for flower classification and localization. It was trained on 224×224 images, and it outputs 10 numbers: outputs 0 to 4 are sent through the softmax activation function, and this gives the class probabilities (one per class); output 5 is sent through the sigmoid activation function, and this gives the objectness score; outputs 6 and 7 represent the bounding box’s center coordinates, and they also go through a sigmoid activation function to ensure they range from 0 to 1; lastly, outputs 8 and 9 represent the bounding box’s height and width, and they do not go through any activation function to allow the bounding boxes to extend beyond the borders of the image. We can now convert the CNN’s dense layers to convolutional layers. In fact, we don’t even need to retrain it; we can just copy the weights from the dense layers to the convolutional layers! Alternatively, we could have converted the CNN into an FCN before training.

Now suppose the last convolutional layer before the output layer (also called the bottleneck layer) outputs 7×7 feature maps when the network is fed a 224×224 image (see the left side of [Figure 14-26](#)). If we feed the FCN a 448×448 image (see the right side of [Figure 14-26](#)), the bottleneck layer will now output 14×14 feature maps.³² Since the dense output layer was replaced by a convolutional layer using 10 filters of size 7×7 , with “valid” padding and stride 1, the output will be composed of 10 feature maps, each of size 8×8 (since $14 - 7 + 1 = 8$). In other words, the FCN will process the whole image only once, and it will output an 8×8 grid where each cell contains 10 numbers (5 class probabilities, 1 objectness score, and 4 bounding box coordinates). It’s exactly like taking the original CNN and sliding it across the image using 8 steps per row and 8 steps per column. To visualize this, imagine chopping the original image into a 14×14 grid, then sliding a 7×7 window across this grid; there will be $8 \times 8 = 64$ possible locations for the window, hence 8×8 predictions. However, the FCN approach is *much* more efficient, since the network only looks at the image once. In fact, *You Only Look Once* (YOLO) is the name of a very popular object detection architecture, which we’ll look at next.

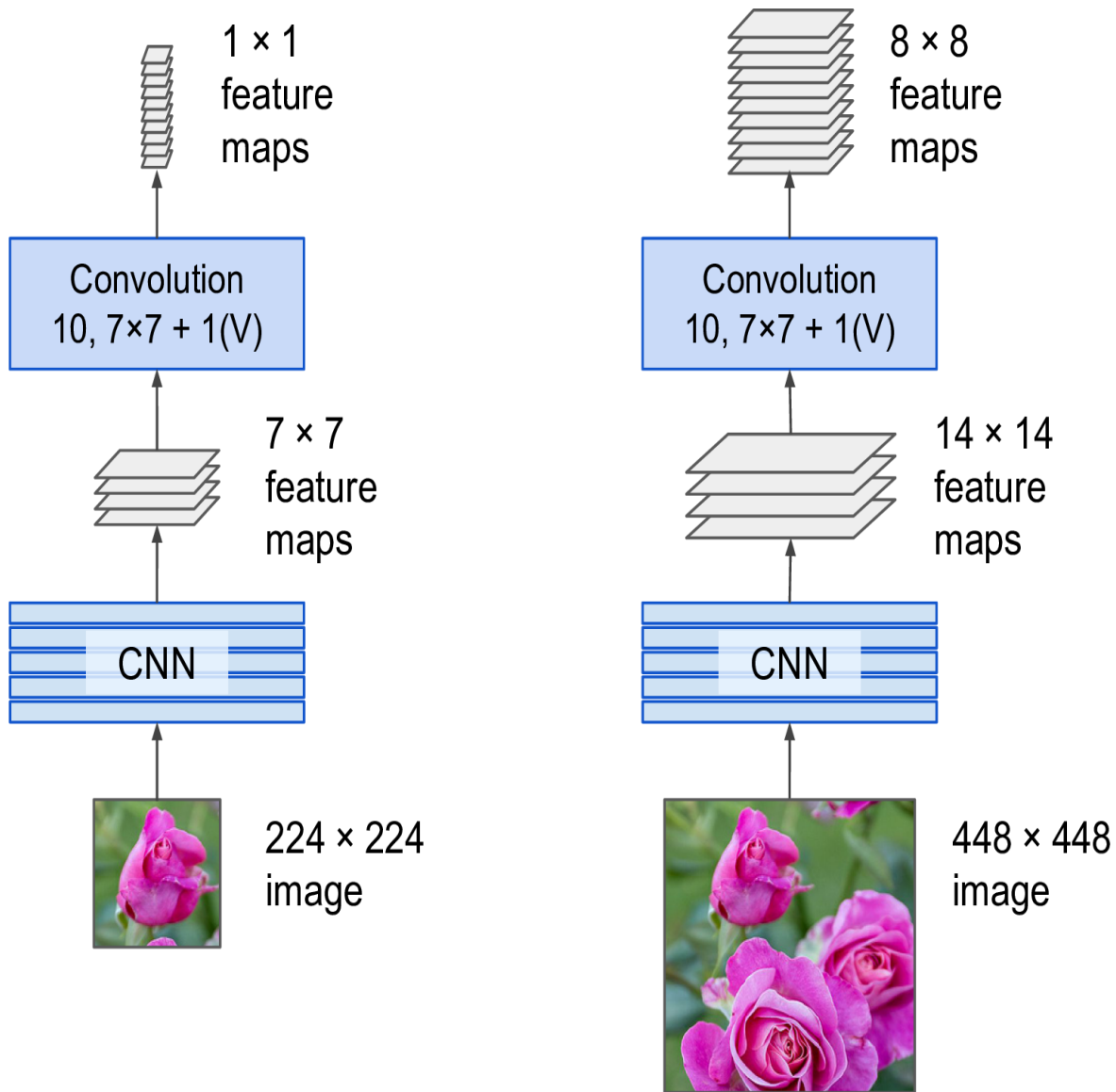


Figure 14-26. The same fully convolutional network processing a small image (left) and a large one (right)

You Only Look Once (YOLO)

YOLO is a fast and accurate object detection architecture proposed by Joseph Redmon et al. in a [2015 paper](#)³³. It is so fast that it can run in real time on a video, as seen in Redmon's [demo](#). YOLO's architecture is quite similar to the one we just discussed, but with a few important differences:

- For each grid cell, YOLO only considers objects whose bounding box center lies within that cell. The bounding box coordinates are relative to that cell, where (0, 0) means the top left of that cell and (1, 1) means the bottom right. However, the bounding box's height and width may extend well beyond the cell.
- It outputs two bounding boxes for each grid cell (instead of just one), which allows the model to handle cases where two objects are so close to each other that their bounding box centers lie within the same cell. Each bounding box also comes with its own objectness score.
- YOLO also outputs a class probability distribution for each grid cell, predicting 20 class probabilities per grid cell since YOLO was trained on the PASCAL VOC dataset, which contains 20 classes. This produces a coarse

class probability map. Note that the model predicts one class probability distribution per grid cell, not per bounding box. However, it's possible to estimate class probabilities for each bounding box during post-processing, by measuring how well each bounding box matches each class in the class probability map. For example, imagine a picture of a person standing in front of a car. There will be two bounding boxes: one large horizontal one for the car, and a smaller vertical one for the person. These bounding boxes may have their centers within the same grid cell. So how can we tell which class should be assigned to each bounding box? Well, the class probability map will contain a large region where the “car” class is dominant, and inside it there will be a smaller region where the “person” class is dominant. Hopefully, the car’s bounding box will roughly match the “car” region, while the person’s bounding box will roughly match the “person” region: this will allow the correct class to be assigned to each bounding box.

YOLO was originally developed using Darknet, an open source Deep Learning framework initially developed in C by Joseph Redmon, but it was soon ported to TensorFlow, Keras, PyTorch, and more. It was continuously improved over the years, with YOLOv2, YOLOv3 and YOLO9000 (again by Joseph Redmon et al.), YOLOv4 (by Alexey Bochkovskiy et al.), YOLOv5 (by Glenn Jocher), and PP-YOLO (by Xiang Long et al.). Each version brought some impressive improvements in speed and accuracy, using a variety of techniques. For example, YOLOv3 boosted accuracy in part thanks to *anchor priors*, exploiting the fact that some bounding box shapes are more likely than others, depending on the class (e.g., people tend to have vertical bounding boxes, while cars usually don’t). They also increased the number of bounding boxes per grid cell, they trained on different datasets with many more classes (up to 9,000 classes organized in a hierarchy in the case of YOLO9000), they added skip connections to recover some of the spatial resolution that is lost in the CNN (we will discuss this shortly, when we look at semantic segmentation), and much more. There are also many variants of these models, such as YOLOv4-tiny, which is optimized to be trained on less powerful machines, and which can run extremely fast (at over 1,000 frames per second!), but with a slightly lower *mean Average Precision* (mAP).

MEAN AVERAGE PRECISION (MAP)

A very common metric used in object detection tasks is the *mean Average Precision* (mAP). “Mean Average” sounds a bit redundant, doesn’t it? To understand this metric, let’s go back to two classification metrics we discussed in [Chapter 3](#): precision and recall. Remember the trade-off: the higher the recall, the lower the precision. You can visualize this in a precision/recall curve (see [Figure 3-6](#)). To summarize this curve into a single number, we could compute its area under the curve (AUC). But note that the precision/recall curve may contain a few sections where precision actually goes up when recall increases, especially at low recall values (you can see this at the top left of [Figure 3-6](#)). This is one of the motivations for the mAP metric.

Suppose the classifier has 90% precision at 10% recall, but 96% precision at 20% recall. There’s really no trade-off here: it simply makes more sense to use the classifier at 20% recall rather than at 10% recall, as you will get both higher recall and higher precision. So instead of looking at the precision *at* 10% recall, we should really be looking at the *maximum* precision that the classifier can offer with *at least* 10% recall. It would be 96%, not 90%. Therefore, one way to get a fair idea of the model’s performance is to compute the maximum precision you can get with at least 0% recall, then 10% recall, 20%, and so on up to 100%, and then calculate the mean of these maximum precisions. This is called the *Average Precision* (AP) metric. Now when there are more than two classes, we can compute the AP for each class, and then compute the mean AP (mAP). That’s it!

In an object detection system, there is an additional level of complexity: what if the system detected the correct class, but at the wrong location (i.e., the bounding box is completely off)? Surely we should not count this as a positive prediction. One approach is to define an IOU threshold: for example, we may consider that a prediction is correct only if the IOU is greater than, say, 0.5, and the predicted class is correct. The corresponding mAP is generally noted mAP@0.5 (or mAP@50%, or sometimes just AP₅₀). In some competitions (such as the PASCAL VOC challenge), this is what is done. In others (such as the COCO competition), the mAP is computed for different IOU thresholds (0.50, 0.55, 0.60, ..., 0.95), and the final metric is the mean of all these mAPs (noted mAP@[.50:.95] or mAP@[.50:0.05:.95]). Yes, that’s a mean mean average.

Many object detection models are available on TensorFlow Hub, often with pretrained weights, such as YOLOv5³⁴, SSD³⁵, Faster R-CNN,³⁶ or EfficientDet³⁷.

SSD and EfficientDet are “look once” detection models, similar to YOLO. EfficientDet is based on the EfficientNet convolutional architecture. Faster R-CNN is more complex: the image first goes through a CNN, then the output is passed to a *Region Proposal Network* (RPN) that proposes bounding boxes that are most likely to contain an object, and a classifier is then run for each bounding box, based on the cropped output of the CNN. The best place to start using these models is TensorFlow Hub’s excellent [object detection tutorial](#).

So far, we’ve only considered detecting objects in single images. But what about videos? Objects must not only be detected in each frame, they must also be tracked over time. Let’s take a quick look at object tracking now.

Object Tracking

Object tracking is a challenging task since objects move, they may grow or shrink as they get closer or further away from the camera, their appearance may change as they turn around or move to different lighting conditions or backgrounds, they may be temporarily occluded by other objects, and so on.

One of the most popular *object tracking* systems is DeepSORT³⁸. It is based on a combination of classical algorithms and Deep Learning:

- It uses *Kalman Filters* to estimate the most likely current position of an object given prior detections, and assuming that objects tend to move at a constant speed.
- It also uses a Deep Learning model to measure the resemblance between new detections and existing tracked objects.
- Lastly, it uses the *Hungarian algorithm* to map new detections to existing tracked objects (or to new tracked objects): this algorithm efficiently finds the combination of mappings that minimizes the distance between the detections and the predicted positions of tracked objects, while also minimizing the appearance discrepancy.

For example, imagine a red ball that just bounced on a blue ball. Based on the previous positions of the balls, the Kalman Filter will predict that the balls will go through each other: indeed, it assumes that objects move at a constant speed, so it will not expect the bounce. If the Hungarian algorithm only considered positions, then it would happily map the new detections to the wrong balls, as if they just went through each other and swapped colors. But thanks to the resemblance measure, the Hungarian algorithm will notice the problem. Assuming the balls are not too similar, the algorithm will map the new detections to the correct balls.

TIP

There are a few DeepSORT implementations available on GitHub. For example, this TensorFlow implementation of YOLOv4 + DeepSORT: <https://github.com/theAIGuysCode/yolov4-deepsort>.

So far we have located objects using bounding boxes. This is often sufficient, but sometimes you need to locate objects with much more precision, for example to remove the background behind a person during a videoconference call. Let’s see how to go down to the pixel level.

Semantic Segmentation

In *semantic segmentation*, each pixel is classified according to the class of the object it belongs to (e.g., road, car, pedestrian, building, etc.), as shown in [Figure 14-27](#). Note that different objects of the same class are *not* distinguished. For example, all the bicycles on the right side of the segmented image end up as one big lump of pixels. The main difficulty in this task is that when images go through a regular CNN, they gradually lose their

spatial resolution (due to the layers with strides greater than 1); so, a regular CNN may end up knowing that there's a person somewhere in the bottom left of the image, but it will not be much more precise than that.

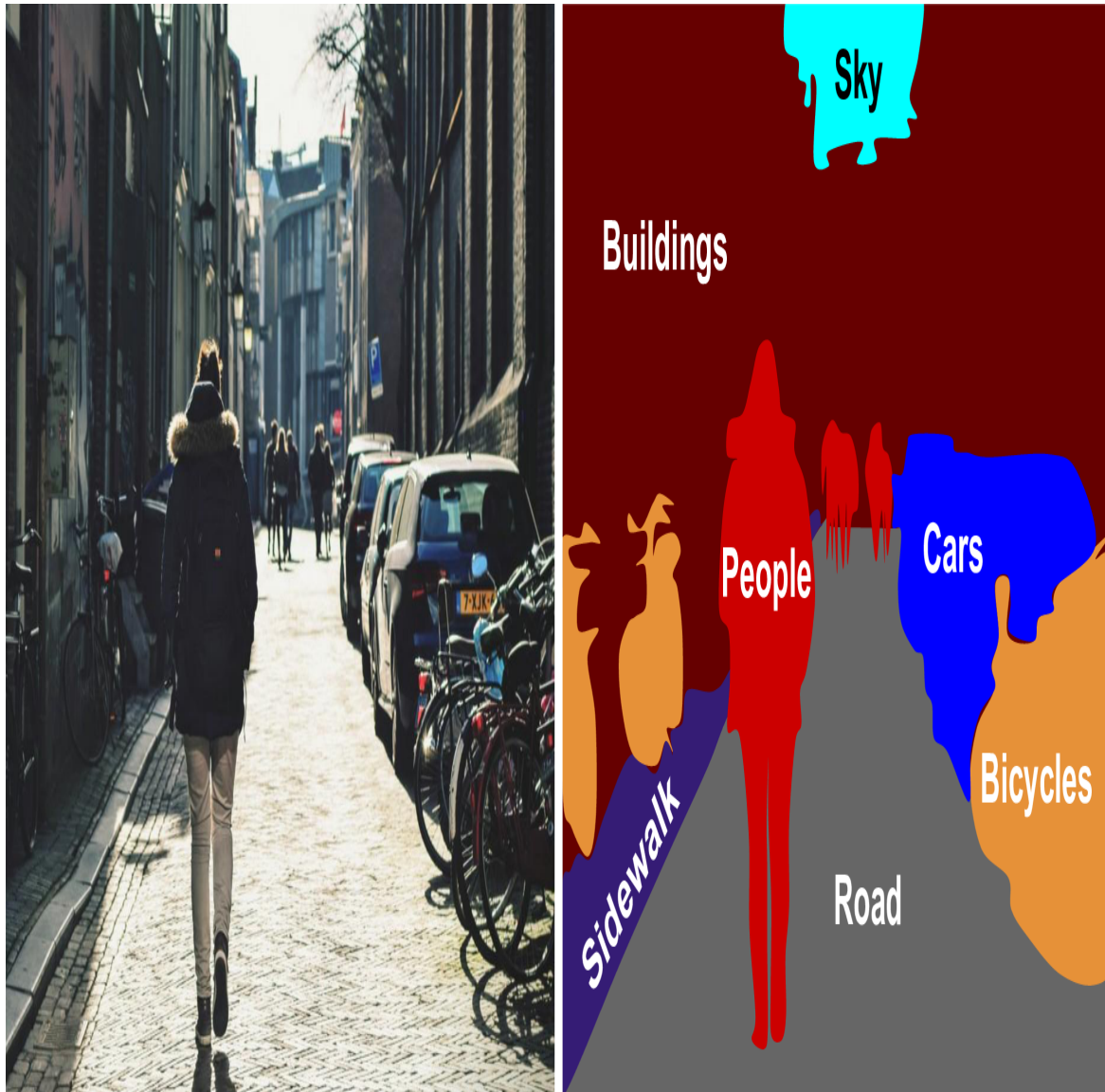


Figure 14-27. Semantic segmentation

Just like for object detection, there are many different approaches to tackle this problem, some quite complex. However, a fairly simple solution was proposed in the 2015 paper by Jonathan Long et al. we discussed earlier, on Fully Convolutional Networks. The authors start by taking a pretrained CNN and turning it into an FCN. The CNN applies an overall stride of 32 to the input image (i.e., if you add up all the strides greater than 1), meaning the last layer outputs feature maps that are 32 times smaller than the input image. This is clearly too coarse, so they add a single *upsampling layer* that multiplies the resolution by 32.

There are several solutions available for upsampling (increasing the size of an image), such as bilinear interpolation, but that only works reasonably well up to $\times 4$ or $\times 8$. Instead, they use a *transposed convolutional layer*:³⁹ it is equivalent to first stretching the image by inserting empty rows and columns (full of zeros), then performing a regular convolution (see Figure 14-28). Alternatively, some people prefer to think of it as a regular convolutional layer that uses fractional strides (e.g., $1/2$ in Figure 14-28). The transposed convolutional layer can

be initialized to perform something close to linear interpolation, but since it is a trainable layer, it will learn to do better during training. In Keras, you can use the Conv2DTranspose layer.

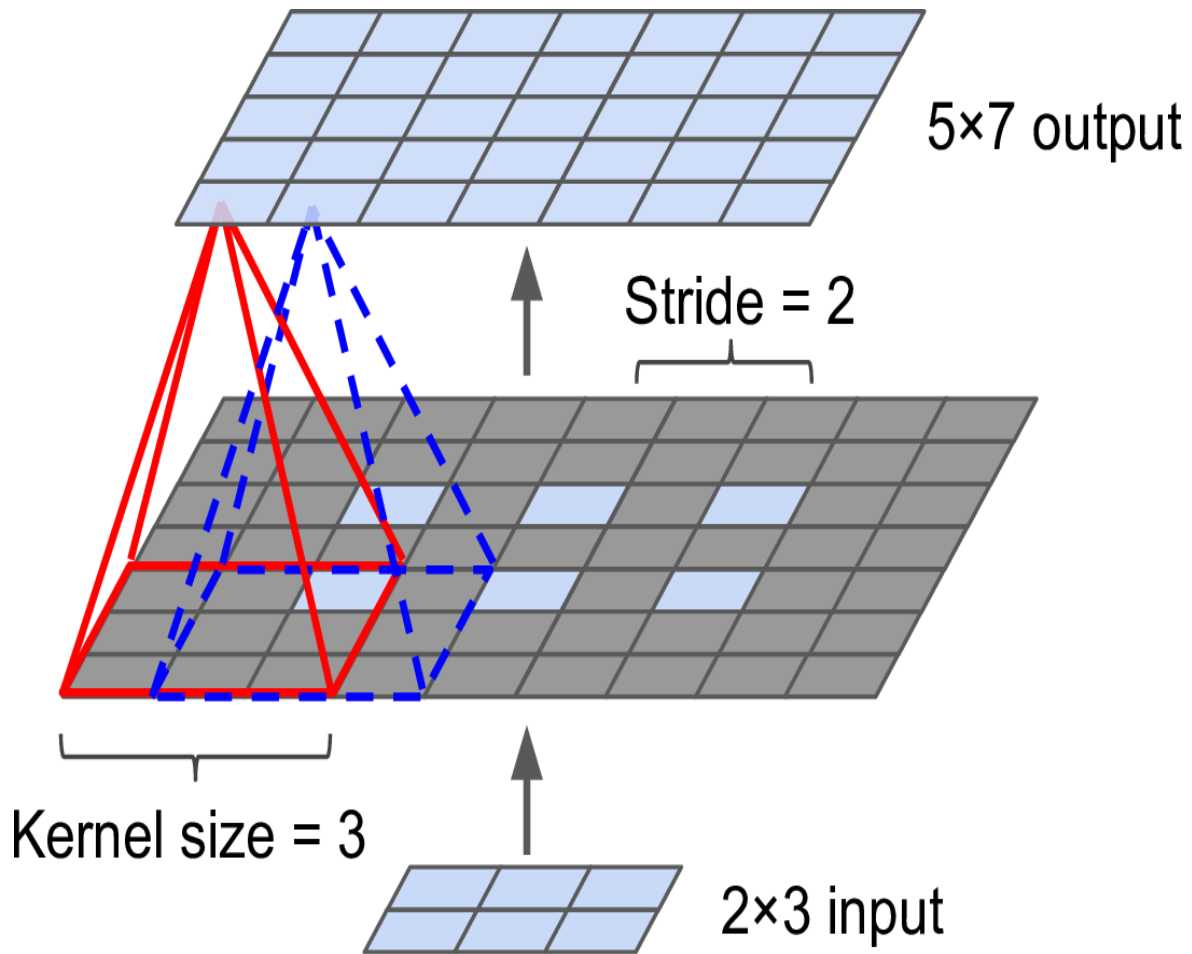


Figure 14-28. Upsampling using a transposed convolutional layer

NOTE

In a transposed convolutional layer, the stride defines how much the input will be stretched, not the size of the filter steps, so the larger the stride, the larger the output (unlike for convolutional layers or pooling layers).

OTHER KERAS CONVOLUTION LAYERS

Keras also offers a few other kinds of convolutional layers:

`tf.keras.layers.Conv1D`

A convolutional layer for 1D inputs, such as time series or text (sequences of letters or words), as we will see in [Chapter 15](#).

`tf.keras.layers.Conv3D`

A convolutional layer for 3D inputs, such as 3D PET scans.

`dilation_rate`

Setting the `dilation_rate` hyperparameter of any convolutional layer to a value of 2 or more creates an *à-trous convolutional layer* (“à trous” is French for “with holes”). This is equivalent to using a regular convolutional layer with a filter dilated by inserting rows and columns of zeros (i.e., holes). For example, a 1×3 filter equal to `[[1, 2, 3]]` may be dilated with a *dilation rate* of 4, resulting in a *dilated filter* of `[[1, 0, 0, 0, 2, 0, 0, 0, 3]]`. This lets the convolutional layer have a larger receptive field at no computational price and using no extra parameters.

Using transposed convolutional layers for upsampling is OK, but still too imprecise. To do better, the authors added skip connections from lower layers: for example, they upsampled the output image by a factor of 2 (instead of 32), and they added the output of a lower layer that had this double resolution. Then they upsampled the result by a factor of 16, leading to a total upsampling factor of 32 (see [Figure 14-29](#)). This recovered some of the spatial resolution that was lost in earlier pooling layers. In their best architecture, they used a second similar skip connection to recover even finer details from an even lower layer. In short, the output of the original CNN goes through the following extra steps: upsample $\times 2$, add the output of a lower layer (of the appropriate scale), upsample $\times 2$, add the output of an even lower layer, and finally upsample $\times 8$. It is even possible to scale up beyond the size of the original image: this can be used to increase the resolution of an image, which is a technique called *super-resolution*.

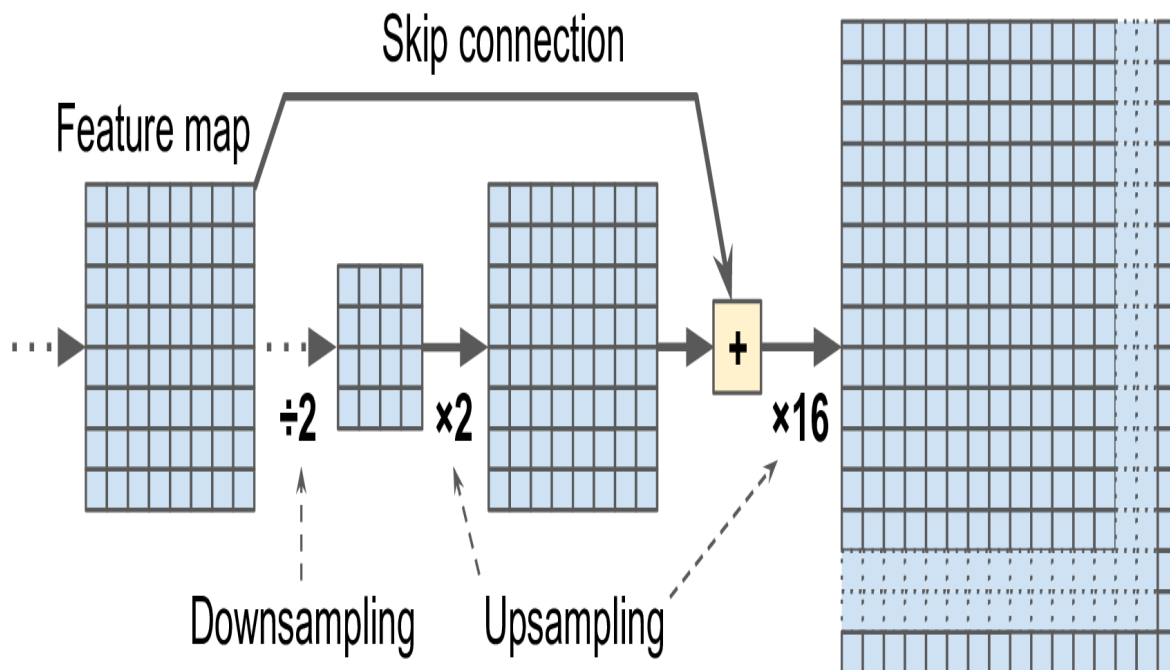


Figure 14-29. Skip layers recover some spatial resolution from lower layers

Instance segmentation is similar to semantic segmentation, but instead of merging all objects of the same class into one big lump, each object is distinguished from the others (e.g., it identifies each individual bicycle). For example the *Mask R-CNN* architecture, proposed in a [2017 paper](#)⁴⁰ by Kaiming He et al., extends the Faster R-CNN model by additionally producing a pixel mask for each bounding box. So not only do you get a bounding box around each object, with a set of estimated class probabilities, but you also get a pixel mask that locates pixels in the bounding box that belong to the object. This model is available on TensorFlow Hub, pretrained on the COCO 2017 dataset. But the field is moving fast, so if you want to try the latest and greatest models, please check out the state-of-the-art section on <https://paperswithcode.com/>.

As you can see, the field of Deep Computer Vision is vast and moving fast, with all sorts of architectures popping out every year. Almost all of them are based on convolutional neural networks, but since 2020 another neural net architecture has entered the computer vision space: Transformers (which we will discuss in [Chapter 16](#)). The progress made over the last decade has been astounding, and researchers are now focusing on harder and harder problems, such as *adversarial learning* (which attempts to make the network more resistant to images designed to fool it), *explainability* (understanding why the network makes a specific classification), realistic *image generation* (which we will come back to in [Chapter 17](#)), *single-shot learning* (a system that can recognize an object after it has seen it just once), predicting the next frames in a video, combining text and image tasks, and more.

Now on to the next chapter, where we will look at how to process sequential data such as time series using recurrent neural networks and convolutional neural networks.

Exercises

1. What are the advantages of a CNN over a fully connected DNN for image classification?
2. Consider a CNN composed of three convolutional layers, each with 3×3 kernels, a stride of 2, and “same” padding. The lowest layer outputs 100 feature maps, the middle one outputs 200, and the top one outputs 400. The input images are RGB images of 200×300 pixels.

What is the total number of parameters in the CNN? If we are using 32-bit floats, at least how much RAM will this network require when making a prediction for a single instance? What about when training on a mini-batch of 50 images?

3. If your GPU runs out of memory while training a CNN, what are five things you could try to solve the problem?
4. Why would you want to add a max pooling layer rather than a convolutional layer with the same stride?
5. When would you want to add a local response normalization layer?
6. Can you name the main innovations in AlexNet, compared to LeNet-5? What about the main innovations in GoogLeNet, ResNet, SEnet, Xception, and EfficientNet?
7. What is a fully convolutional network? How can you convert a dense layer into a convolutional layer?
8. What is the main technical difficulty of semantic segmentation?
9. Build your own CNN from scratch and try to achieve the highest possible accuracy on MNIST.
10. Use transfer learning for large image classification, going through these steps:
 - a. Create a training set containing at least 100 images per class. For example, you could classify your own pictures based on the location (beach, mountain, city, etc.), or alternatively you can use an existing dataset (e.g., from TensorFlow Datasets).
 - b. Split it into a training set, a validation set, and a test set.
 - c. Build the input pipeline, apply the appropriate preprocessing operations, and optionally add data augmentation.
 - d. Fine-tune a pretrained model on this dataset.
11. Go through TensorFlow's [Style Transfer tutorial](#). It is a fun way to generate art using Deep Learning.

Solutions to these exercises are available at the end of this chapter's notebook, at <https://homl.info/colab3>.

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- 1 David H. Hubel, "Single Unit Activity in Striate Cortex of Unrestrained Cats," *The Journal of Physiology* 147 (1959): 226–238.
 - 2 David H. Hubel and Torsten N. Wiesel, "Receptive Fields of Single Neurons in the Cat's Striate Cortex," *The Journal of Physiology* 148 (1959): 574–591.
 - 3 David H. Hubel and Torsten N. Wiesel, "Receptive Fields and Functional Architecture of Monkey Striate Cortex," *The Journal of Physiology* 195 (1968): 215–243.
 - 4 Kunihiro Fukushima, "Neocognitron: A Self-Organizing Neural Network Model for a Mechanism of Pattern Recognition Unaffected by Shift in Position," *Biological Cybernetics* 36 (1980): 193–202.
 - 5 Yann LeCun et al., "Gradient-Based Learning Applied to Document Recognition," *Proceedings of the IEEE* 86, no. 11 (1998): 2278–2324.
 - 6 A convolution is a mathematical operation that slides one function over another and measures the integral of their pointwise multiplication. It has deep connections with the Fourier transform and the Laplace transform and is heavily used in signal processing. Convolutional layers actually use cross-correlations, which are very similar to convolutions (see <https://homl.info/76> for more details).
 - 7 To produce the same size outputs, a fully connected layer would need $200 \times 150 \times 100$ neurons, each connected to all $150 \times 100 \times 3$ inputs. It would have $200 \times 150 \times 100 \times (150 \times 100 \times 3 + 1) \approx 135$ billion parameters!
 - 8 In the international system of units (SI), $1 \text{ MB} = 1,000 \text{ KB} = 1,000 \times 1,000 \text{ bytes} = 1,000 \times 1,000 \times 8 \text{ bits}$. And $1 \text{ MiB} = 1,024 \text{ KiB} = 1,024 \times 1,024 \text{ bytes}$. So $12 \text{ MB} \approx 11.44 \text{ MiB}$.
 - 9 Other kernels we've discussed so far had weights, but pooling kernels do not: they are just stateless sliding windows.
 - 10 Yann LeCun et al., "Gradient-Based Learning Applied to Document Recognition," *Proceedings of the IEEE* 86, no. 11 (1998): 2278–2324.
 - 11 Alex Krizhevsky et al., "ImageNet Classification with Deep Convolutional Neural Networks," *Proceedings of the 25th International Conference on Neural Information Processing Systems* 1 (2012): 1097–1105.
 - 12 Matthew D. Zeiler and Rob Fergus, "Visualizing and Understanding Convolutional Networks," *Proceedings of the European Conference on Computer Vision* (2014): 818–833.
 - 13 Christian Szegedy et al., "Going Deeper with Convolutions," *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition* (2015): 1–9.
 - 14 In the 2010 movie *Inception*, the characters keep going deeper and deeper into multiple layers of dreams; hence the name of these modules.

- 15 Karen Simonyan and Andrew Zisserman, “Very Deep Convolutional Networks for Large-Scale Image Recognition,” arXiv preprint arXiv:1409.1556 (2014).
- 16 Kaiming He et al., “Deep Residual Learning for Image Recognition,” arXiv preprint arXiv:1512.03385 (2015).
- 17 It is a common practice when describing a neural network to count only layers with parameters.
- 18 Christian Szegedy et al., “Inception–v4, Inception-ResNet and the Impact of Residual Connections on Learning,” arXiv preprint arXiv:1602.07261 (2016).
- 19 François Chollet, “Xception: Deep Learning with Depthwise Separable Convolutions,” arXiv preprint arXiv:1610.02357 (2016).
- 20 This name can sometimes be ambiguous, since spatially separable convolutions are often called “separable convolutions” as well.
- 21 Jie Hu et al., “Squeeze-and-Excitation Networks,” *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition* (2018): 7132–7141.
- 22 Saining Xie et al., “Aggregated Residual Transformations for Deep Neural Networks,” arXiv preprint arXiv:1611.05431 (2016).
- 23 Gao Huang et al., “Densely Connected Convolutional Networks,” arXiv preprint arXiv:1608.06993 (2016).
- 24 Andrew G. Howard et al., “MobileNets: Efficient Convolutional Neural Networks for Mobile Vision Applications,” arXiv preprint arxiv:1704.04861 (2017).
- 25 Chien-Yao Wang et al., “CSPNet: A New Backbone that can Enhance Learning Capability of CNN,” arXiv preprint arXiv:1911.11929 (2019).
- 26 Mingxing Tan and Quoc V. Le, “EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks,” arXiv preprint arXiv:1905.11946 (2019).
- 27 A 92-core AMD EPYC CPU with IBPB, 1.7T of RAM, and an Nvidia Tesla A100 GPU.
- 28 In the ImageNet dataset, each image is mapped to a word in the **WordNet dataset**: the class ID is just a WordNet ID.
- 29 Adriana Kovashka et al., “Crowdsourcing in Computer Vision,” *Foundations and Trends in Computer Graphics and Vision* 10, no. 3 (2014): 177–243.
- 30 Jonathan Long et al., “Fully Convolutional Networks for Semantic Segmentation,” *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition* (2015): 3431–3440.
- 31 There is one small exception: a convolutional layer using “valid” padding will complain if the input size is smaller than the kernel size.
- 32 This assumes we used only “same” padding in the network: indeed, “valid” padding would reduce the size of the feature maps. Moreover, 448 can be neatly divided by 2 several times until we reach 7, without any rounding error. If any layer uses a different stride than 1 or 2, then there may be some rounding error, so again the feature maps may end up being smaller.
- 33 Joseph Redmon et al., “You Only Look Once: Unified, Real-Time Object Detection,” *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition* (2016): 779–788.
- 34 You can find YOLOv3, YOLOv4, and their tiny variants in the TensorFlow Models project at <https://homl.info/yolof>.
- 35 Wei Liu et al., “SSD: Single Shot Multibox Detector,” *Proceedings of the 14th European Conference on Computer Vision* 1 (2016): 21–37.
- 36 Shaoqing Ren et al., “Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks,” *Proceedings of the 28th International Conference on Neural Information Processing Systems* 1 (2015): 91–99.
- 37 Mingxing Tan et al., “EfficientDet: Scalable and Efficient Object Detection,” arXiv preprint arXiv:1911.09070 (2019).
- 38 Nicolai Wojke et al., “Simple Online and Realtime Tracking with a Deep Association Metric,” arXiv preprint arXiv:1703.07402 (2017).
- 39 This type of layer is sometimes referred to as a *deconvolution layer*, but it does *not* perform what mathematicians call a deconvolution, so this name should be avoided.
- 40 Kaiming He et al., “Mask R-CNN,” arXiv preprint arXiv:1703.06870 (2017).

Chapter 15. Processing Sequences Using RNNs and CNNs

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This will be the 15th chapter of the final book. The GitHub repo is <https://github.com/ageron/handson-ml3>.

If you have comments about how we might improve the content and/or examples in this book, or if you notice missing material within this chapter, please reach out to the editor at mcronin@oreilly.com.

Predicting the future is something you do all the time, whether you are finishing a friend’s sentence or anticipating the smell of coffee at breakfast. In this chapter we will discuss recurrent neural networks (RNNs)—a class of nets that can predict the future (well, up to a point). RNNs can analyze time series data, such as the number of daily active users on your website, the hourly temperature in your city, your home’s daily power consumption, or the trajectories of nearby cars, and more. Once an RNN learns past patterns in the data, it is able to use its knowledge to forecast the future, assuming of course that past patterns still hold in the future.

More generally, RNNs can work on sequences of arbitrary lengths, rather than on fixed-sized inputs. For example, they can take sentences, documents, or audio samples as input, making them extremely useful for natural language processing applications such as automatic translation or speech-to-text.

In this chapter, we will first go through the fundamental concepts underlying RNNs and how to train them using backpropagation through time. Then, we will use them to forecast a time series. Along the way, we will look at the popular ARMA family of models, often used to forecast time series, and use them as baselines to compare with our RNNs. After that, we'll explore the two main difficulties that RNNs face:

- Unstable gradients (discussed in [Chapter 11](#)), which can be alleviated using various techniques, including *recurrent dropout* and *recurrent layer normalization*.
- A (very) limited short-term memory, which can be extended using LSTM and GRU cells.

RNNs are not the only types of neural networks capable of handling sequential data. For small sequences, a regular dense network can do the trick, and for very long sequences, such as audio samples or text, convolutional neural networks can actually work quite well too. We will discuss both of these possibilities, and we will finish this chapter by implementing a *WaveNet*—a CNN architecture capable of handling sequences of tens of thousands of time steps. Let's get started!

Recurrent Neurons and Layers

Up to now we have focused on feedforward neural networks, where the activations flow only in one direction, from the input layer to the output layer. A recurrent neural network looks very much like a feedforward neural network, except it also has connections pointing backward.

Let's look at the simplest possible RNN, composed of one neuron receiving inputs, producing an output, and sending that output back to itself, as shown in [Figure 15-1](#) (left). At each *time step* t (also called a *frame*), this *recurrent neuron* receives the inputs $\mathbf{x}_{(t)}$ as well as its own output from the previous time step, $\hat{y}_{(t-1)}$. Since there is no previous output at the first time step, it is generally set to 0. We can represent this tiny network against the time axis, as shown in [Figure 15-1](#) (right). This is called *unrolling the network*

through time (it's the same recurrent neuron represented once per time step).

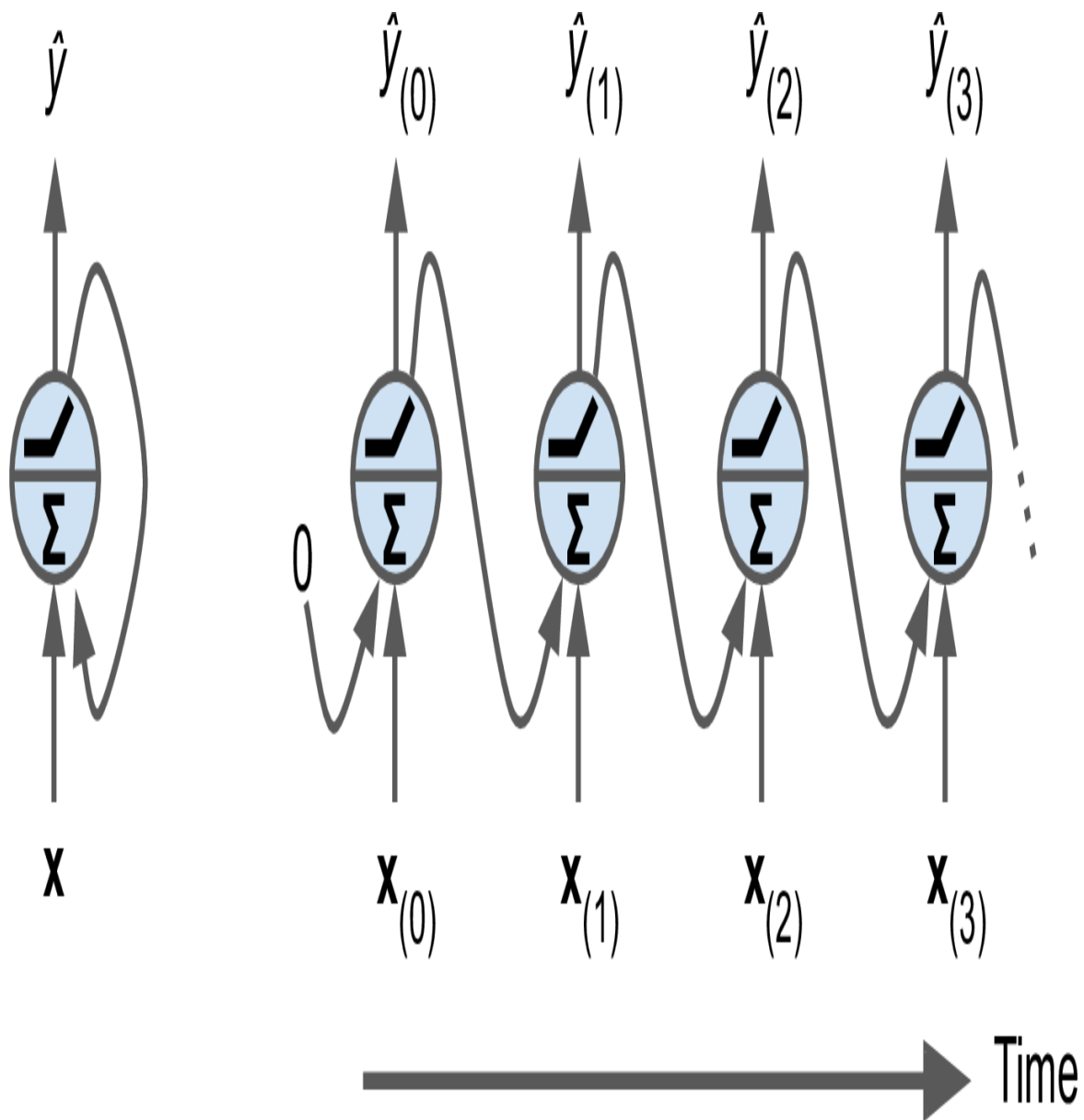


Figure 15-1. A recurrent neuron (left) unrolled through time (right)

You can easily create a layer of recurrent neurons. At each time step t , every neuron receives both the input vector $\mathbf{x}_{(t)}$ and the output vector from the previous time step $\hat{\mathbf{y}}_{(t-1)}$, as shown in [Figure 15-2](#). Note that both the inputs and outputs are now vectors (when there was just a single neuron, the output was a scalar).

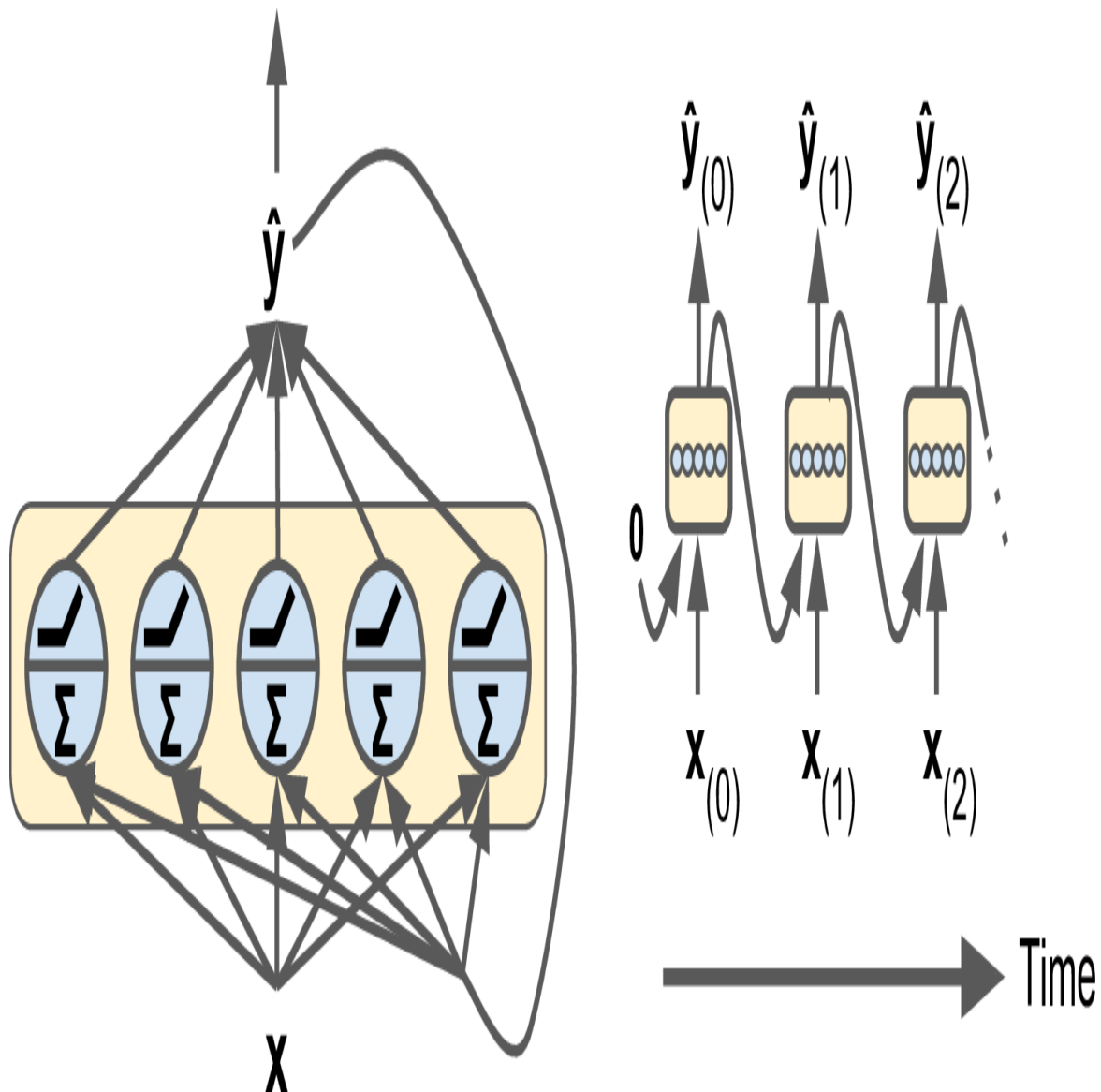


Figure 15-2. A layer of recurrent neurons (left) unrolled through time (right)

Each recurrent neuron has two sets of weights: one for the inputs $x_{(t)}$ and the other for the outputs of the previous time step, $\hat{y}_{(t-1)}$. Let's call these weight vectors w_x and $w_{\hat{y}}$. If we consider the whole recurrent layer instead of just one recurrent neuron, we can place all the weight vectors in two weight matrices: W_x and $W_{\hat{y}}$.

The output vector of the whole recurrent layer can then be computed pretty much as you might expect, as shown in [Equation 15-1](#), where b is the bias vector and $\phi(\cdot)$ is the activation function (e.g., ReLU¹).

Equation 15-1. Output of a recurrent layer for a single instance

$$\hat{\mathbf{y}}_{(t)} = \phi(\mathbf{W}_x^\top \mathbf{x}_{(t)} + \mathbf{W}_{\hat{\mathbf{y}}}^\top \hat{\mathbf{y}}_{(t-1)} + \mathbf{b})$$

Just as with feedforward neural networks, we can compute a recurrent layer's output in one shot for an entire mini-batch by placing all the inputs at time step t into an input matrix $\mathbf{X}_{(t)}$ (see [Equation 15-2](#)).

Equation 15-2. Outputs of a layer of recurrent neurons for all instances in a mini-batch

$$\begin{aligned}\hat{\mathbf{Y}}_{(t)} &= \phi(\mathbf{X}_{(t)} \mathbf{W}_x + \hat{\mathbf{Y}}_{(t-1)} \mathbf{W}_{\hat{\mathbf{y}}} + \mathbf{b}) \\ &= \phi\left(\begin{bmatrix} \mathbf{X}_{(t)} & \hat{\mathbf{Y}}_{(t-1)} \end{bmatrix} \mathbf{W} + \mathbf{b}\right) \text{ with } \mathbf{W} = \begin{bmatrix} \mathbf{W}_x \\ \mathbf{W}_{\hat{\mathbf{y}}} \end{bmatrix}\end{aligned}$$

In this equation:

- $\hat{\mathbf{Y}}_{(t)}$ is an $m \times n_{\text{neurons}}$ matrix containing the layer's outputs at time step t for each instance in the mini-batch (m is the number of instances in the mini-batch and n_{neurons} is the number of neurons).
- $\mathbf{X}_{(t)}$ is an $m \times n_{\text{inputs}}$ matrix containing the inputs for all instances (n_{inputs} is the number of input features).
- $\mathbf{W}_x$ is an $n_{\text{inputs}} \times n_{\text{neurons}}$ matrix containing the connection weights for the inputs of the current time step.
- $\mathbf{W}_{\hat{\mathbf{y}}}$ is an $n_{\text{neurons}} \times n_{\text{neurons}}$ matrix containing the connection weights for the outputs of the previous time step.
- $\mathbf{b}$ is a vector of size n_{neurons} containing each neuron's bias term.
- The weight matrices $\mathbf{W}_x$ and $\mathbf{W}_{\hat{\mathbf{y}}}$ are often concatenated vertically into a single weight matrix $\mathbf{W}$ of shape $(n_{\text{inputs}} + n_{\text{neurons}}) \times n_{\text{neurons}}$ (see the second line of [Equation 15-2](#)).
- The notation $\begin{bmatrix} \mathbf{X}_{(t)} & \hat{\mathbf{Y}}_{(t-1)} \end{bmatrix}$ represents the horizontal concatenation of the matrices $\mathbf{X}_{(t)}$ and $\hat{\mathbf{Y}}_{(t-1)}$.